

Open APIs
for Open
Minds

FIWARE Data Spaces Foundations and Technical Architecture

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Data Spaces: our vision

Data Spaces are **governed ecosystems** enabling participants to **sustainably create value** by interacting in a **trusted**, **sovereign** and **agile** manner

- Focus on **sustainable creation of value**:
 - Participants need to find the right incentives to invest (RoI)
 - Value-creation should be fair and balanced
- **Trusted**: participants can interact with confidence
- **Sovereign**: participants can exercise control over their data and manage their users autonomously
- **Agile**: participants can offer and consume services easily, in an effective and seamless “plug & play” manner



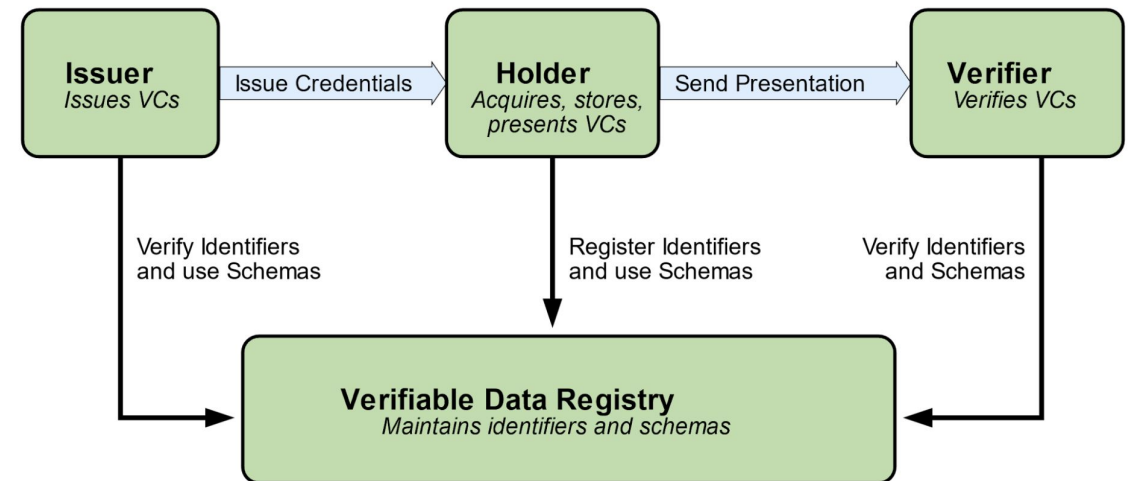
Data Spaces requires an enough wide vision to succeed

- Data Spaces cannot become “Open Data with steroids”
- It is not just about organizations publishing data, it is also about enabling application providers to offer services
 - for accessing, processing, visualizing and interpreting data
 - not just through APIs serving applications but GUIs directly serving end users
- It is not just about organizations consuming plain data, it is also about enabling users of organizations to consume application services
 - natural persons in the first place (employees, customers)
 - also devices or software agents owning identity
- A narrow vision introduces severe barriers for adoption:
 - “I’m a provider of an app which allow visitors display best routes in a map and destination managers the analysis of visitors preferences and behavior ... are you telling me that I can just provide raw data?”
 - “If our city wishes to offer smart routes to visitors ... are you telling me I have to get the JSON-LD of a calculated route and deal with how to display it on maps for visitors?”



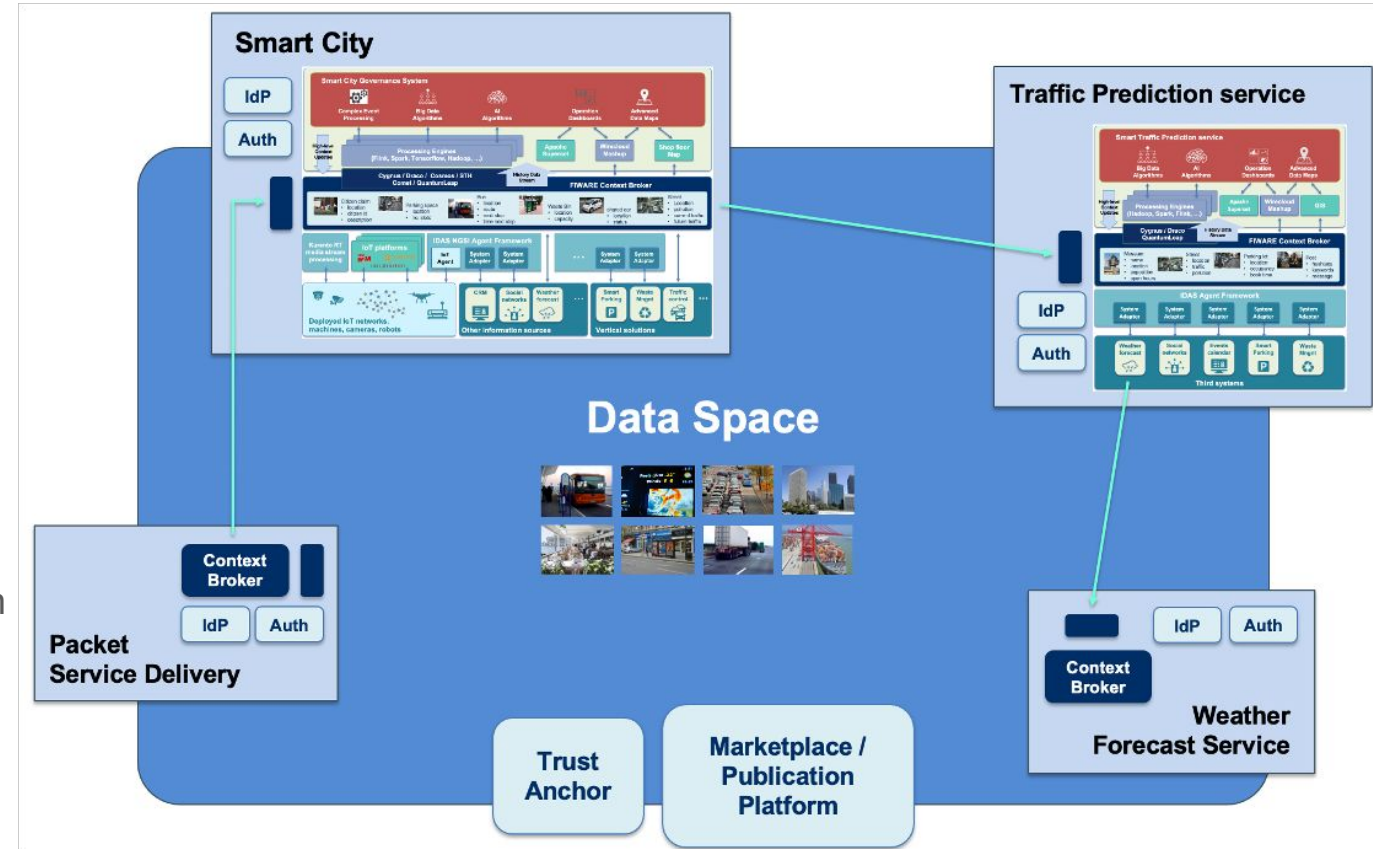
Some key concepts: Users, Decentralized IAM

- Users within organizations participating in a data space can consume Products offered by other participants:
 - users can be natural persons (employees, customers), devices, or software agents (including AI agents)
 - each organization requires managing its own users: user identity management has to be decentralized
- DIDs (Decentralized Identifiers) and Verifiable Credentials (VCs) will be used to describe organizations and users:
 - model analogous to the one everyone has used for years with physical cards, thus easier to understand
 - enables decentralization and supports privacy, adding security
- VCs can also be used to provide attestation of certain characteristics of products offered by providers, e.g.:
 - characteristics certified by trusted certification agencies, describing compliance with certain regulations (e.g., GDPR compliance) recommendations (e.g., low carbon emissions) or technical compliance (e.g., NGSI-LD compatible interface).
 - characteristics given by the own service provider describing aspects of the service (e.g., pricing model, access policies, technical interoperability standards supported, etc)



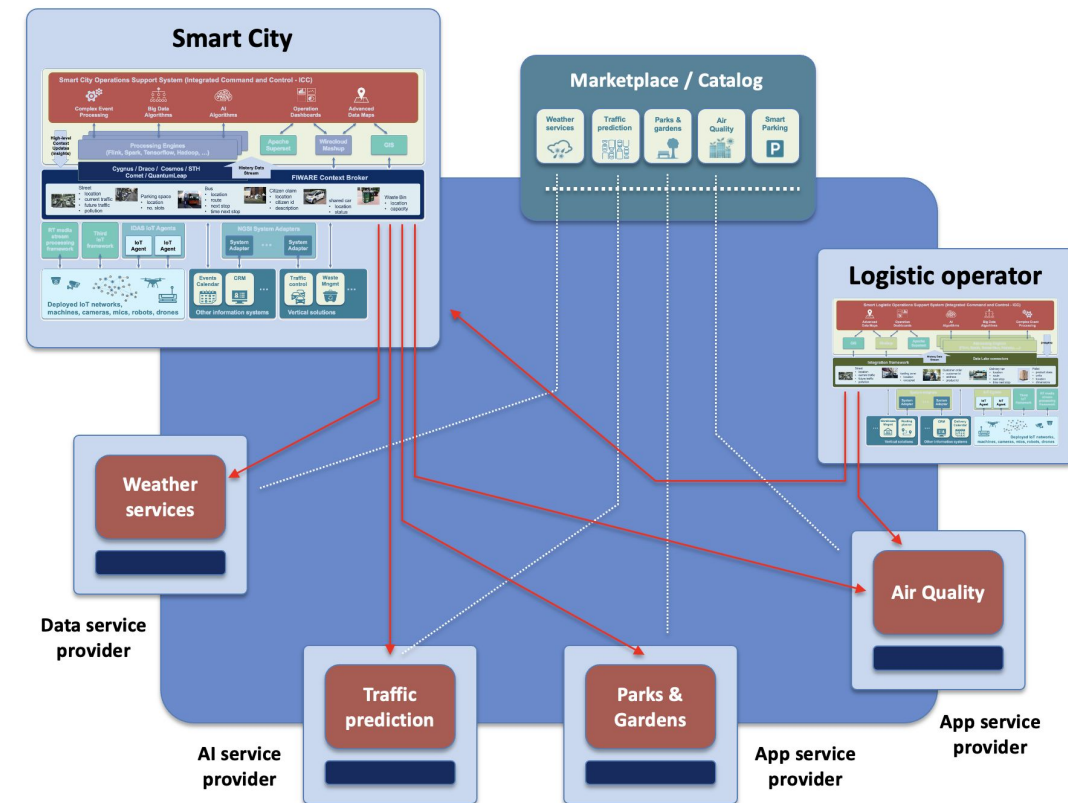
Some key concepts: acquiring rights to use a product

- Inherent to the business logic of products is the definition of authorization policies establishing what users can do based on claims in their VCs:
 - park visitors vs gardeners in connection to a parks&gardens management product
 - premium vs regular customers in connection to a packet service delivery product
- Participants that acquire rights to use a product:
 - become trusted issuers of VCs including claims relevant for accessing product services
 - acquiring organizations decide what VCs to assign to users linked to the organization (employees, customers, devices, software agents)
- Product providers get rid of the complexity of managing identity of users from their customers



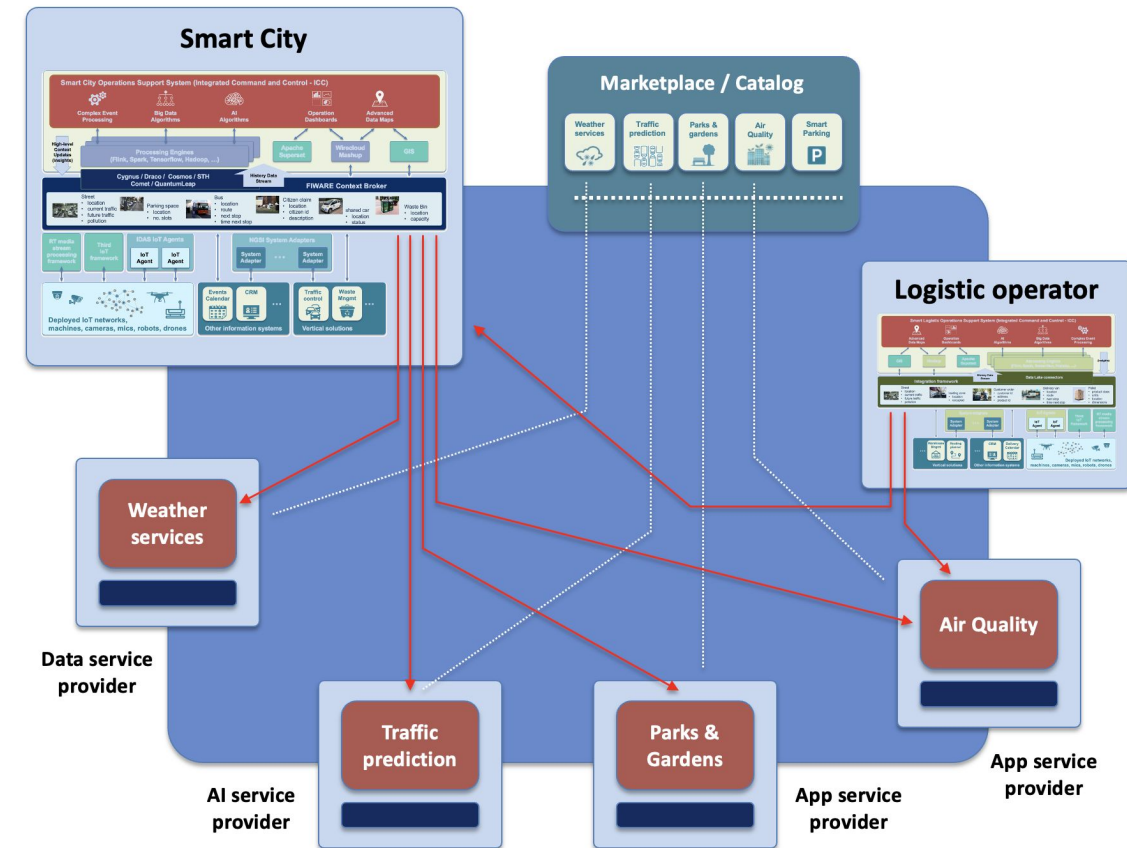
Some fundamental position statements on Data Spaces

- Data Spaces are not just about B2B exchange of datasets !!
 - invocation of services (for data processing, actuation)
 - access to data interpretation/visualization apps
 - users (employees, customers, devices, robots, software agents) linked to consumer organizations need to also access services based on their profile within the organization
- Data Spaces comes with a more powerful vision - they are open innovation ecosystems involving:
 - providers offering products that range from simple services to access datasets to complex applications comprising several services, including data processing services
 - organizations extending their systems architecture via integration of services linked to products offered by third providers, without giving up management of their users
 - users linked to consumer organizations involved as direct consumers of product services
- For any data space to become a truly open ecosystem to which organizations can dynamically join and interact, interoperability standards at multiple levels need to be agreed as part of the governance



Areas where interoperability has to be addressed (2/2)

- **APIs/protocols for exchange** among participants, e.g.:
 - APIs/protocols for invoking near-realtime data access and processing services (e.g., NGSI-LD)
 - APIs/protocols for exchange of files / large datasets (e.g, S3)
- **Data Models:**
 - common vocabularies and semantics
 - common formats for the serialization of data
- Mechanisms for **Publication and discovery** of:
 - product specifications with their characteristics, including associated services and resources
 - offerings around products, including pricing as well as other Terms and Conditions
- Commonly agreed **process to acquire the rights to use products** (systems) offered by any provider:
 - standard API for accessing characteristics of a product and managing the process for the acquisition of rights to use it
 - acquiring the rights to use a product by an organization implies that it becomes a trusted issuer of the credentials services linked to the product require to be accessed



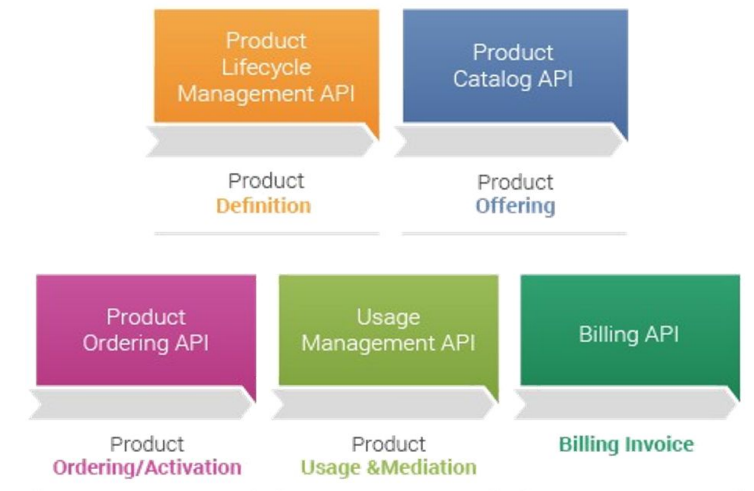
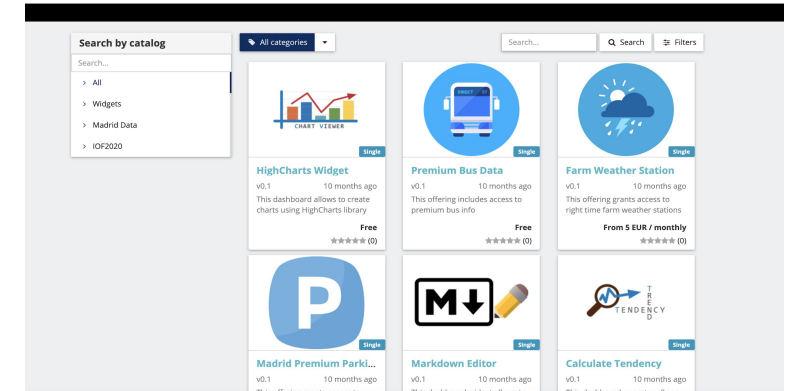
Reference specifications: Trust and Sovereignty

- Decentralized Identity Management and authentication relies on:
 - W3C [DID \(Decentralized Identifiers\)](#) and [Verifiable Credentials \(VC\)](#)
 - OpenID [OID4VC](#) protocols, compatible with [EU DI Wallet Architecture](#):
 - Verifiable Credentials Issuance Protocol: [OID4VCI](#)
 - Verifiable Credentials Exchange Protocol: [OID4VP](#)
- In an open ecosystem allowing organizations to join, a Trust Framework is required bringing:
 - a registry of trusted participants the verifier checks to determine whether the organization signing a VC is a legal entity that complies with ecosystem rules
 - a registry of trusted issuers of VCs the verifier checks to determine whether a valid ecosystem participant is entitled to sign certain kind of rules
 - For both, we rely on [EBSI API](#) specs
 - The onboarding service which brings the mechanisms to check entities are actually legal persons and comply with ecosystem rules may rely on services from [Gaia-X Digital Clearing Houses](#) for authentication
- Authorization framework following PEP-PDP-PIP and PRP/PAP architecture for ABAC (attributes $\Leftrightarrow$ claims in VCs), and adopting [ODRL](#) as Policy Definition Language (with support to [Gaia-X ODRL VC profile](#))



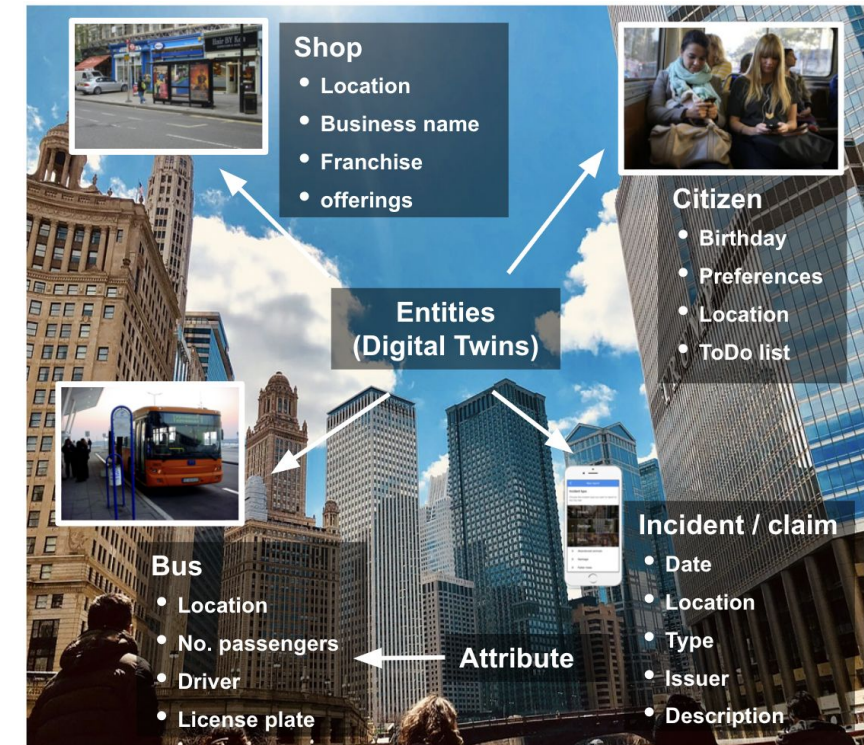
Reference specifications: Value creation

- Creating value out of data and services, ultimately products, being offered is the ultimate goal in data spaces. This follows basically requires to:
 - Describe data, services, resources, products, offerings in an interoperable manner
 - Include data and service publication services to discover offerings facilitating connection of providers and consumers
 - Support contract negotiation peer-to-peer or through value-added services such as independent marketplaces
- [TM Forum Open APIs](#) bring the basis for managing offerings and support contract negotiation via marketplaces or through connectors:
 - easing integration with [DOME](#)
 - compatible with [Eclipse Dataspace Protocol](#) (Catalog and Contract Negotiation)
 - supporting monetization
- It would be desirable that descriptions will also available through [DCAT v3](#) catalogs at connector level or at data space level (e.g., Global Catalogs or Marketplaces)



Reference specifications: Data Exchange

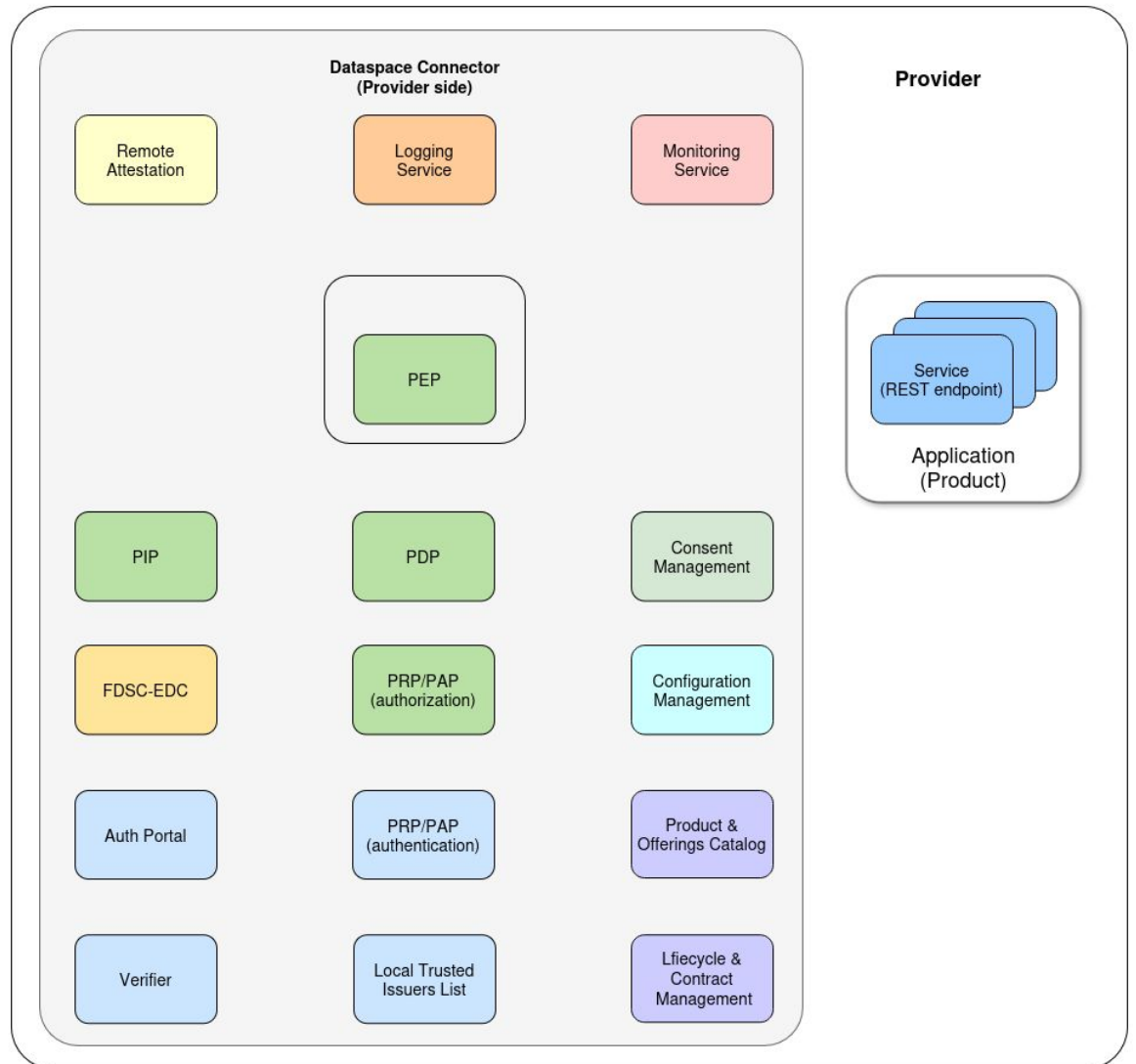
- Providers of data products in data spaces must be able to offer data services at well defined endpoints, where consumers know how to consume their data services.
- This means that all participants in data spaces should 'speak the same language', addressing interoperability at several levels:
 - transport and syntactic level → common APIs
 - semantic level → common data models/vocabularies
- [NGSI-LD](#) is recommended for exchange of digital twin and digital product data; [S3](#) for large data exchange; [MCP/A2A](#) for exchange between AI agents
- Adoption of common data models is encouraged and there are multiple references to consider (ISO/IEC CIM for Energy, SAREF, ...) - the [Smart Data Models initiative](#) brings a hub that solves how models are mapped into JSON, JSON-LD as serialization formats
- Data exchange may be subject to control - [Eclipse Dataspace Protocol \(Transfer Process\)](#) works here
- Still worth highlighting: access to web pages exposing application services (e.g., data visualization) is feasible through the Connector!!



FIWARE Data Space Connector

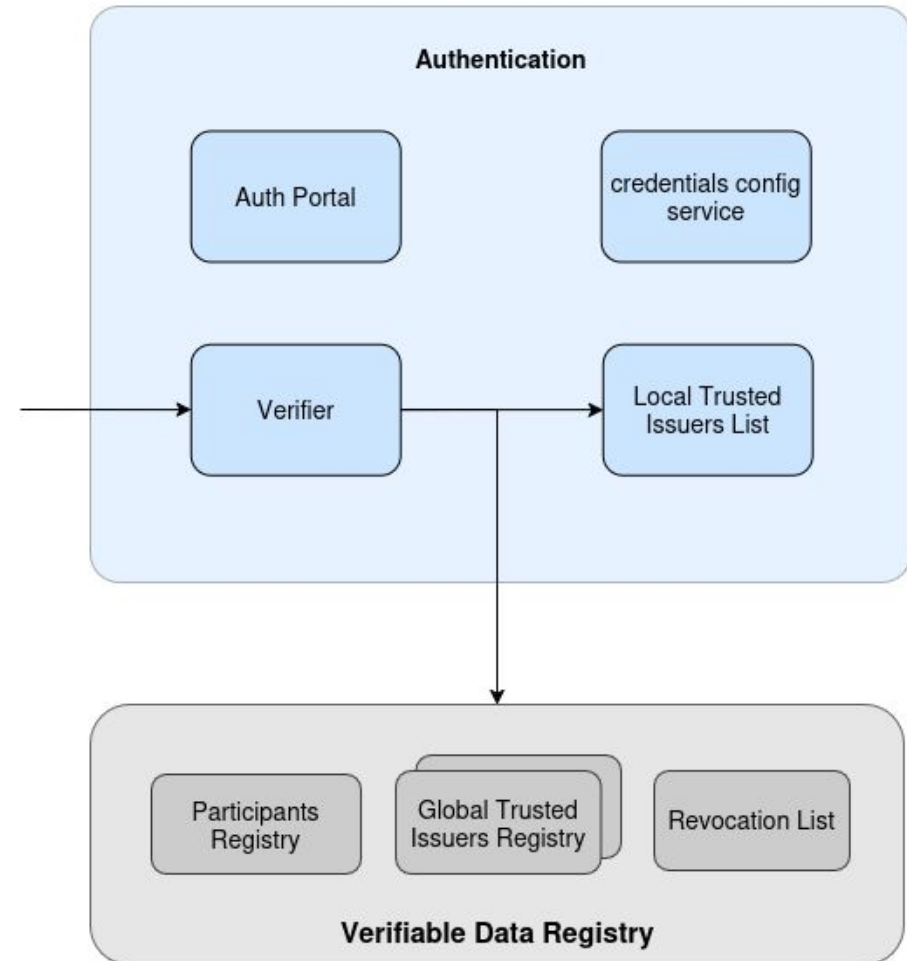
FIWARE Data Space Connector

- The concept of Data Space Connector has evolved to match the idea of an integrated suite of components every organization offering data or services in a data space should deploy
- These components would be deployed and configured in controlled environments (e.g., a Kubernetes cluster) and implement a number of services which may be required for an organization to connect in its role as provider of data and services:
 - Authentication (including the interface to trust services)
 - Authorization (policy enforcement)
 - Catalog Publication
 - Contract Management
 - Logging
 - ...
- This vision is materialized in the FIWARE Data Space Connector framework



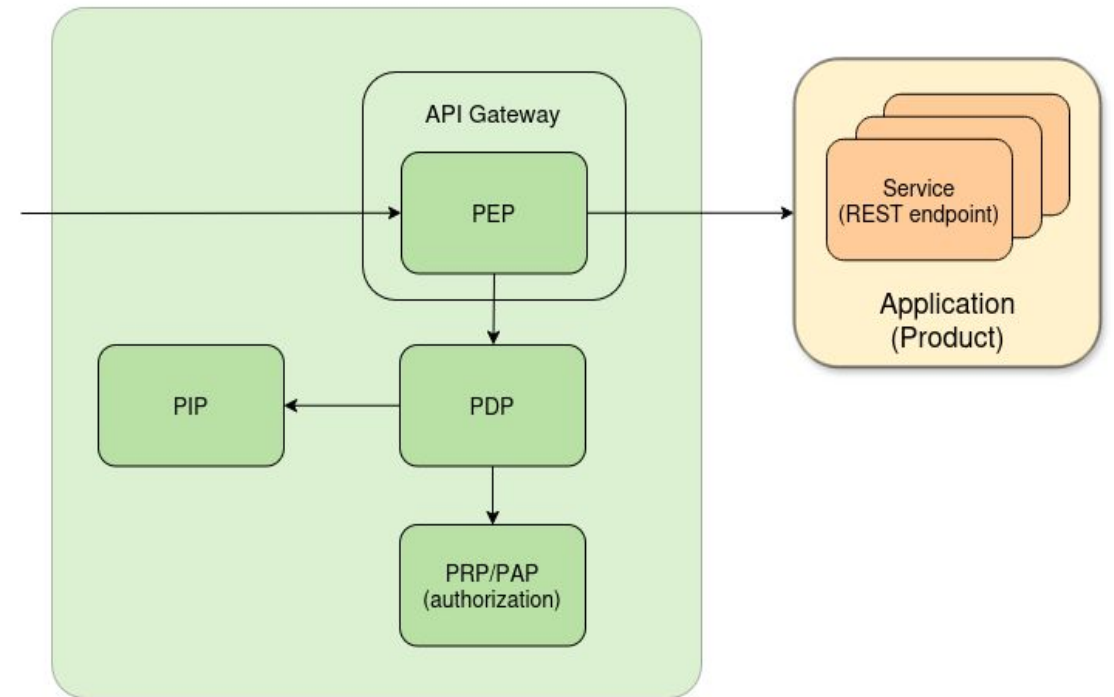
FIWARE Data Space Connector - Authentication

- Verifier to provide authentication capabilities via OID4VP
- Trusted Issuers List for defining credentials trust anchors
- Credentials Config Service provides configuration for the Verifier



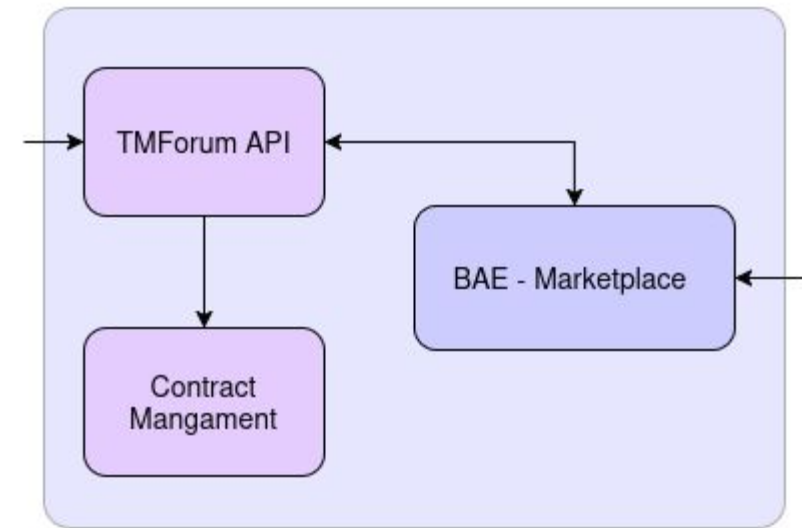
FIWARE Data Space Connector - Authorization

- Policy Enforcement Point(Apisix) as entry point for all external requests towards the Data Space Connector and the provided downstream services
- Policy Decision Point to allow/deny requests to the offered Data/Services
- Policy Administration Point to hold and manage policies, translate and provide them for the PEP
- Policy Information Point to provide additional information for policy evaluation



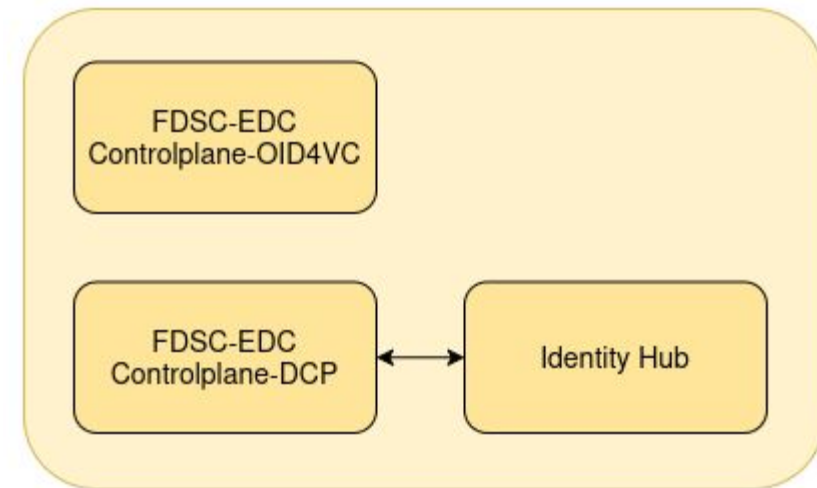
FIWARE Data Space Connector - Contract Management

- TMForum API Implementation as a backend for marketplace functionalities
- Contract Management to manage product flows and integrate them with the connector components(policies, trust anchors, trust configuration)
- BAE Marketplace as GUI for Marketplace functionalities



FIWARE Data Space Connector - EDC Integration

- FDSC-EDC Controlplane for DCP - integration with the Eclipse Dataspace Protocol through the Decentralized Claims Protocol
- FDSC-EDC Controlplane for OID4VC - integration with the Eclipse Dataspace Protocol through OpenId for Verifiable Credential
- Identity Hub to support the Decentralized Claims Protocol



Decentralized Identity and Access Management

Stefan Wiedemann - Senior Software Engineer

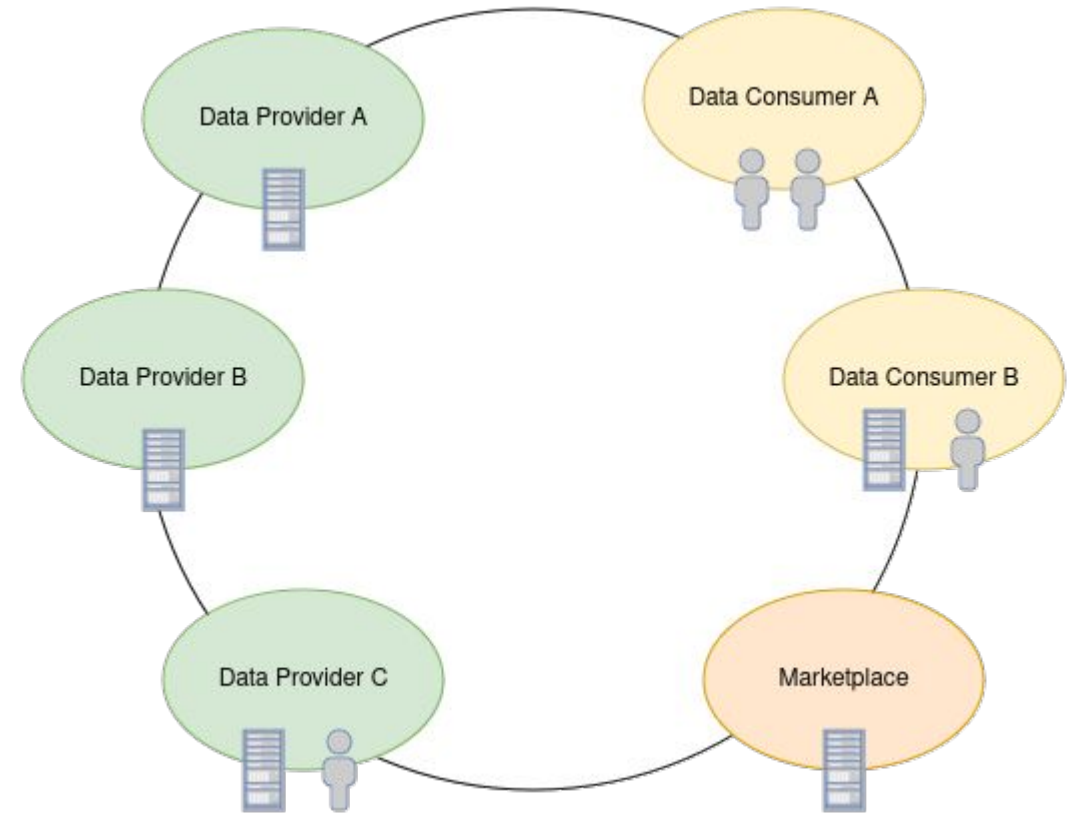


FIWARE Decentralized IAM & DSC

1. Decentralized IAM in Data Spaces - Why?
2. Introduction into Decentralized Identity
 - a. Verifiable Credentials
 - b. Decentralized Identifiers
3. Authentication and Authorization in a FIWARE Data Space
4. Policy Enforcement with ODRL

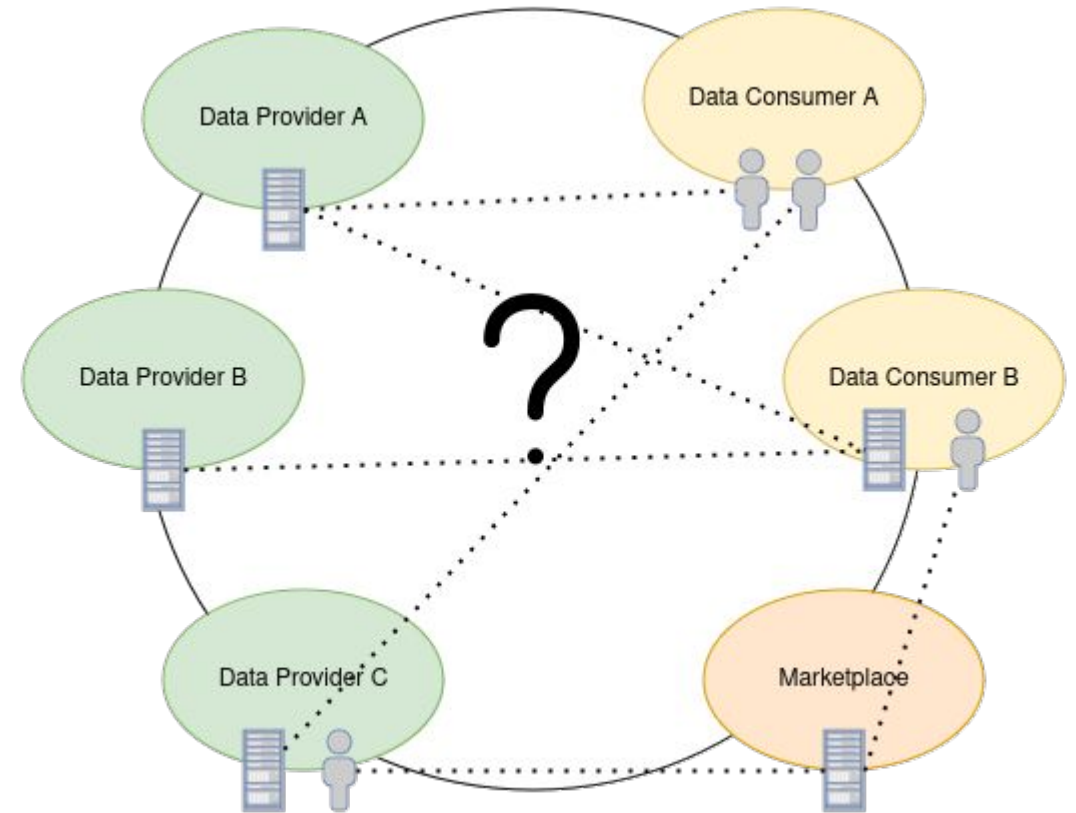
Why?

- A Data Spaces consists of different participants in various roles
- The participating organizations have technical and human actors, connecting each other
- No connection beside the Data Spaces between the participants
- No knowledge about each others actors(human or technical)



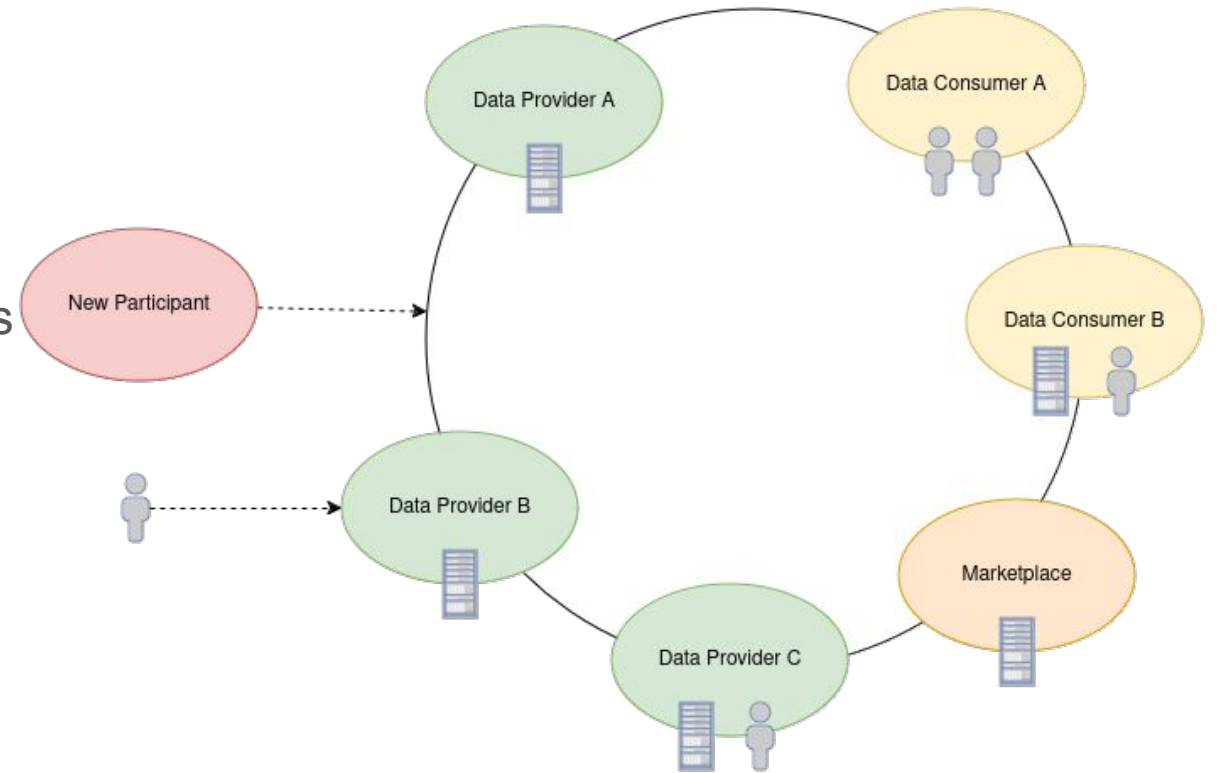
Why?

- How to identify actors?
- How to authorize their actions?
- How to connect them with their organization?



Why?

- New participant joining the Data Space requires a new trust-relationship
- New actor joins one of the participants, needs to be recognized



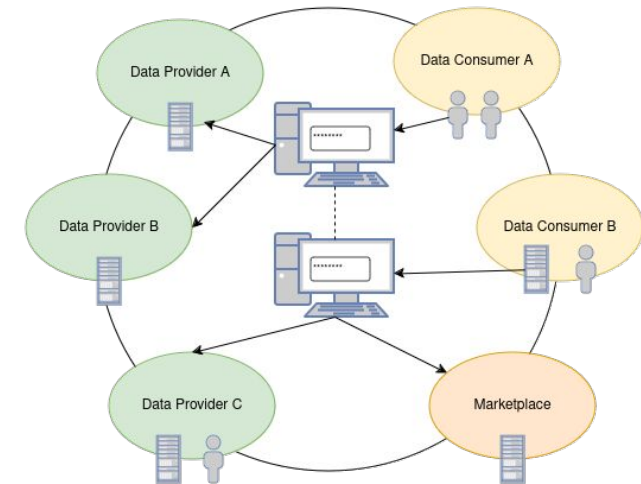
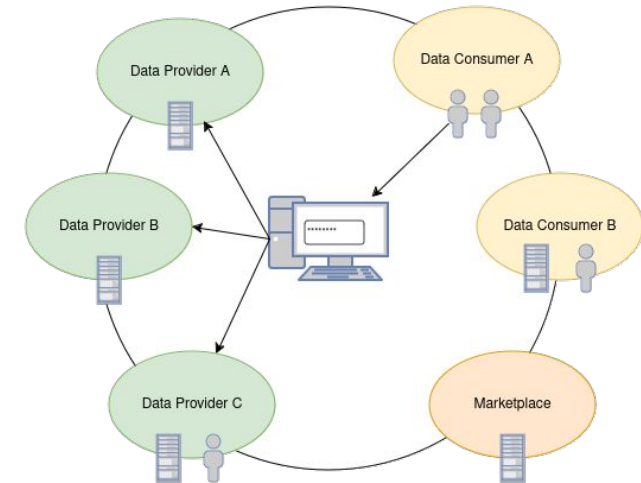
Why?

Central Login:

- Easy to use, but all the power at one single entity
- powerful (user) data sink

Federated Identity:

- Less reliance on a central IDP
- still reliance on “central” providers
- increasing number of providers -> increased complexity

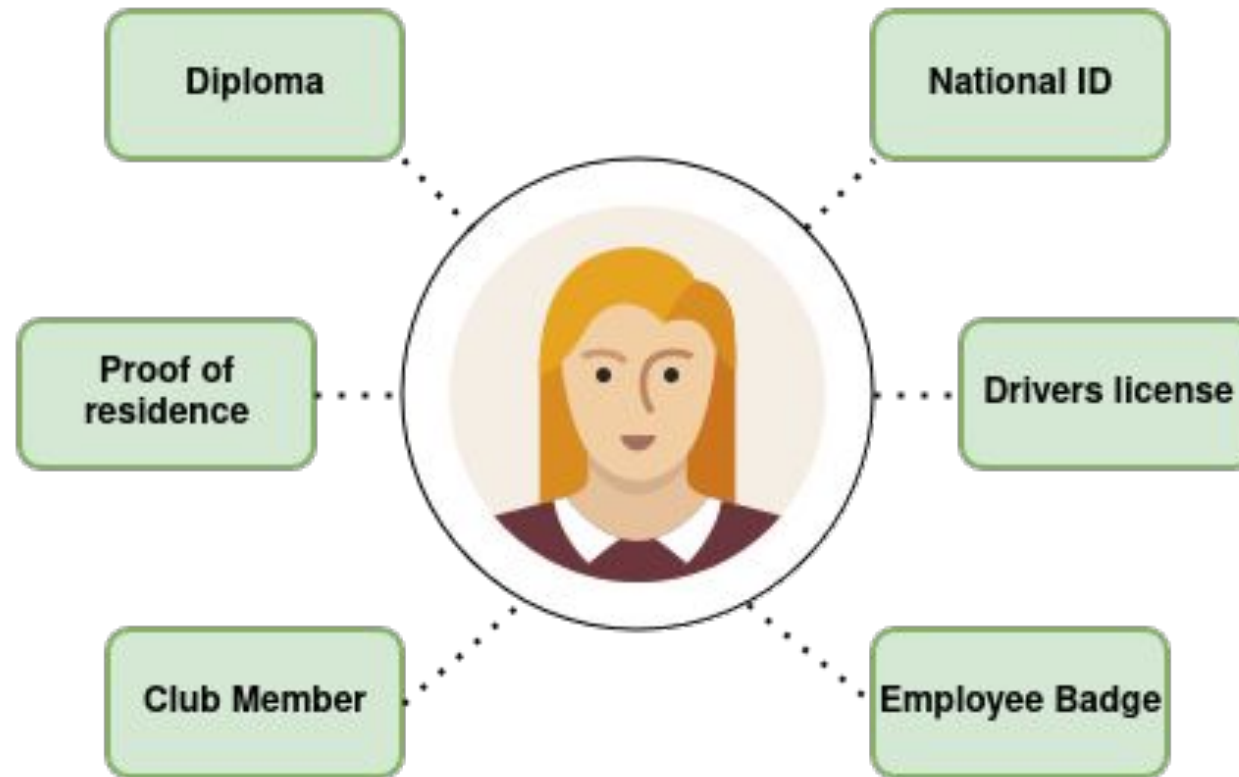


Decentralized Identity

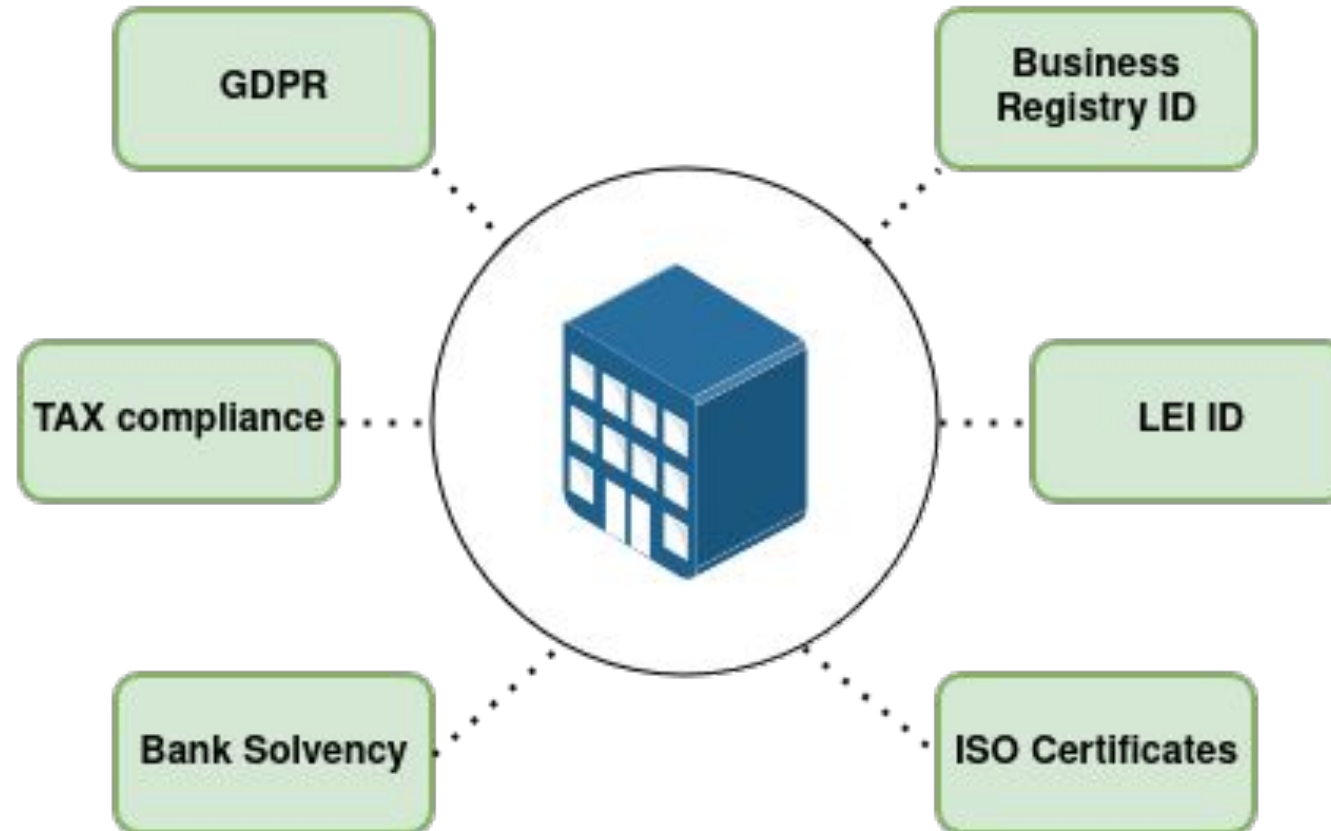
- Copy the good things of the physical world
- Augment it with digital powers
- Preserve the required properties
- Adapt to current legal frameworks, but stay prepared for the future



Identity of a natural person

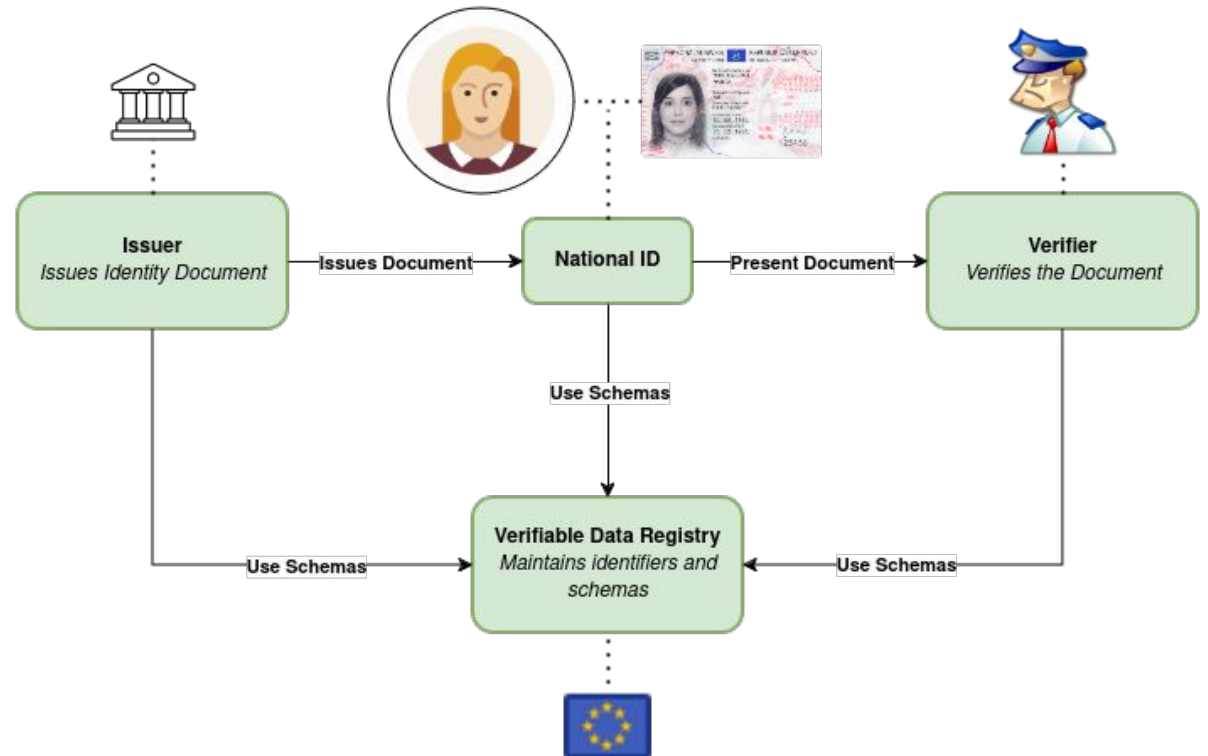


Identity of a legal person



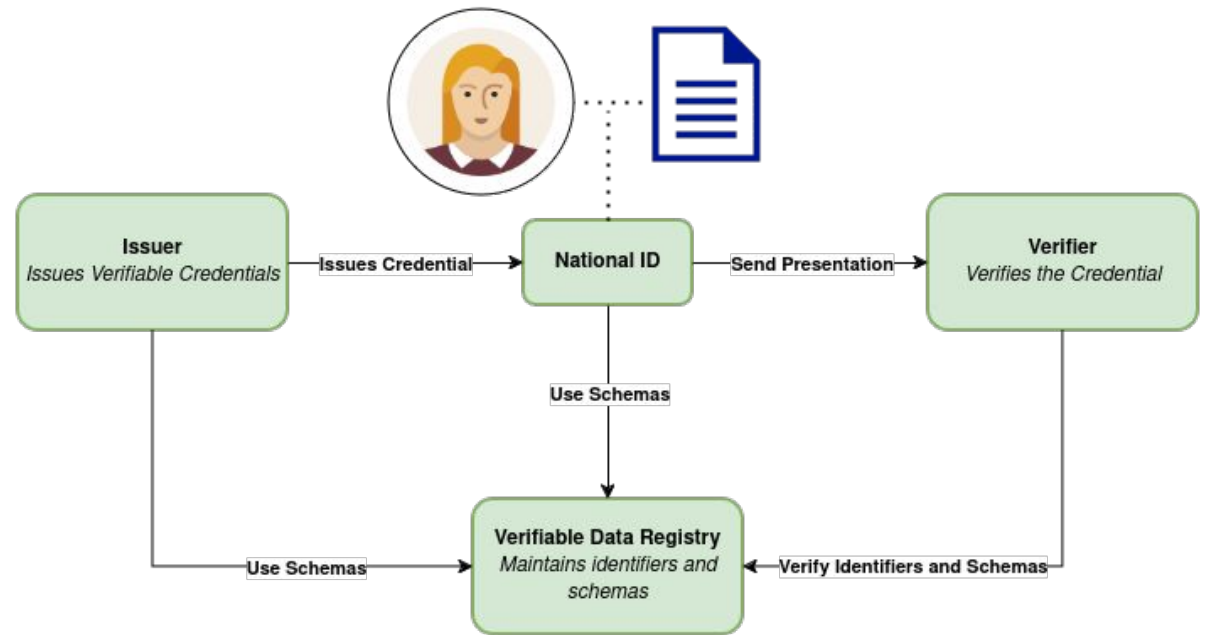
Authentication in the Real World

- Government registered at registry(EU)
- National ID is issued by the government to a citizen
- Follows commonly agreed schemas(defined by the EU)
- Is presented to an authority for authentication
- Verifies authenticity features and checks that its a registered Schema(“National ID”) and from a registered issuer



W3C Verifiable Credentials

- Issuer is registered at the Verifiable Data Registry
- Issuer issues Credential to the holder
- Holder presents the credential to the verifier
- Verifier verifies the signature, schema and checks that issuer and schema are registered at the Registry



W3C Verifiable Credentials

Verifiable Credentials are tamper-proof, digitally signed documents that securely prove claims about an individual or entity, allowing verification without directly involving the issuer.

- Specified by W3C - [VC-Data-Model-2.0](#)
- Defines a basic document structure and the essential features of the credential documents
- Can contain any kind of (json-representable) data
- Tamper-proof, due to digital signatures as part of the credential
- Already adopted by various regulatory and standardization bodys(f.e. [EUDI Wallet Reference Architecture](#), [OpenID4VC](#))

W3C Verifiable Credentials

```
{
  "@context": [
    "https://www.w3.org/2018/credentials/v1",
    "https://ec.europa.eu/eudi/v1"
  ],
  "type": ["VerifiableCredential", "NationalID"],
  "issuanceDate": "2024-01-01T12:00:00Z",
  "expirationDate": "2034-01-01T12:00:00Z",
  "issuer": "did:eu:country-code:issuer-identifier",
  "credentialSubject": {
    "id": "did:eu:country-code:user-identifier",
    "givenName": "Maria",
    "familyName": "Musterfrau",
    "dateOfBirth": "1983-12-01",
    "nationality": "AU"
  },
  "proof": {
    "type": "Ed25519Signature2020",
    "created": "2024-01-01T12:00:00Z",
    "proofPurpose": "assertionMethod",
    "verificationMethod": "did:eu:country-code:issuer-identifier#keys-1",
    "jws": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
  }
}
```

Credential Metadata

Claim(s)

Proof(s)

W3C Verifiable Credentials - Metadata

```
{
  "@context": [
    "https://www.w3.org/2018/credentials/v1",
    "https://ec.europa.eu/eudi/v1"
  ],
  "type": ["VerifiableCredential", "NationalID"],
  "validFrom": "2024-01-01T12:00:00Z",
  "validUntil": "2034-01-01T12:00:00Z",
  "issuer": "did:eu:country-code:issuer-identifier",
  "id": "did:example:ebfeb1f712ebc6f1c276e12ec21"
}
```

- Provides contextual and technical information about the credential itself
- Helps the verifier to understand validity, provenance and usage context
- @context: Provides the context for interpreting the data within the credential, it's often used to link schemas and standards
- type: Defines the types of the credential to help verifiers interpret its purpose
- issuer: Information about the issuing entity. It can be used to verify the document.
- optional metadata like id, names, descriptions, validFrom, validUntil or credentialStatus supported

W3C Verifiable Credentials - Claims

```
{
  "credentialSubject": [{
    "id": "did:eu:country-code:user-identifier",
    "givenName": "Maria",
    "familyName": "Musterfrau",
    "dateOfBirth": "1983-12-01",
    "nationality": "AU",
    "address": {
      "street": "Kärtner Straße",
      "city": "Wien",
      "postalCode": "1010"
    }
  },
  {
    ...
  }
]
```

- Section where the actual information or assertions are stored
- Property “credentialSubject” contains a set of objects, with the specific attributes or data pieces the issuer is attesting about the subject
- The claims usually should follow specific data schemas to ensure consistency and interoperability
- The subjects often contain id's, linking the claims to specific holders. It helps verifiers to confirm that a claim truly applies to the presenting subject.

W3C Verifiable Credentials - Proofs

```
{  
  "proof": {  
    "type": "Ed25519Signature2020",  
    "created": "2024-01-01T12:00:00Z",  
    "proofPurpose": "assertionMethod",  
    "verificationMethod":  
    "did:eu:country-code:issuer-identifier#keys-1",  
    "jws": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."  
  }  
}
```

- The proof contains details about the digital signature on the credential
- “type” contains the proof type. It determines what other fields are required to secure and verify the proof
- “proofPurpose” contains the reason the proof was created for.
- “verificationMethod” provides the information required to verify the proof
- “jws” contains the actual digital signature, which ensures the credential has not been tampered with

W3C Verifiable Presentation

```
{
  "@context": [... ],
  "type": ["VerifiablePresentation"],
  "verifiableCredential": [
    {
      "@context": [... ],
      "type": ["VerifiableCredential", "NationalID"],
      "issuanceDate": "2024-01-01T12:00:00Z",
      "expirationDate": "2034-01-01T12:00:00Z",
      "issuer": "did:eu:country-code:issuer-identifier",
      "credentialSubject": {
        ...
      },
      "proof": {
        ...
      }
    },
    ...
  ],
  "proof": {
    "type": "Ed25519Signature2020",
    "created": "2024-01-01T12:00:00Z",
    "proofPurpose": "assertionMethod",
    "verificationMethod": "...",
    "jws": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
  }
}
```

- A package of one or more Verifiable Credentials, created by the holder, to share specific information with a verifier
- Can combine multiple credentials, which is useful when a verifier required proof of several attributes
- The proof ensures integrity of the data and confirms that it was actually generated by the holder
- Enables the verifier to ensure a credential is provided by the actual holder, by checking the holder information in the credential with the presentations' proof

W3C Verifiable Credentials - Formats

- Various types of Verifiable Credentials are specified at the moment:
 - [jwt-vc](#): represents Verifiable Credentials as JSON Web Tokens. The format leverages established standards like JSON Web Signatures to ensure claims made in the credential can be validated by the verifier.
 - [vc+sd-jwt](#): extension to the JWT standard, that enables privacy preserving VCs by allowing the holders to disclose only specific claims, while maintaining integrity and authenticity of the entire JWT.
 - [mdl](#): specialized structure for representing mobile drivers licenses

Decentralized Identifiers

Decentralized Identifiers(DIDs) are a new type of identifier that enables verifiable, self-sovereign digital identities without relying on a central authority, allowing individual to control their own identities and associated data securely and independent.

- Specified by W3C - [Decentralized Identifiers](#)
- Defines the syntax of an identifier
- Allows the controller of the identity to prove control without requiring permission from anyone else
- DIDs are URIs associated with a document allowing trustable interaction. The document can contain:
 - cryptographic material
 - verification methods
 - services to prove control

Decentralized Identifiers

did:<method-name>:<method-specific-id><additional-components>

- did-prefix: indicates that the given URI is a decentralized identifier
- **method-name**: References a specific DID Method.
- The method specifies:
 - how the **method-specific-id** is generated
 - how the associated DID document is created
 - how the DID can be resolved to the document
 - how the DID document can be used to verify authenticity
- **additional-components**: reference into the DID-document or an external resource, depending on the method

Decentralized Identifiers

- FIWARE Data Space Connector supports 4 methods:
 - [did:key](#)
 - [did:jwk](#)
 - [did:web](#)
 - [did:elsi](#)

did:key:zDnaeTTC781gfJKRjjShQV5sjNxxvgETXSctwfuMXZFmbeSE2

did:jwk:<jwk>

did:web:did-provider.dev.seamware.io

did:elsi:VATDE-309937516

did:key

did:key:zDnaeTTC781gfJKRjjShQV5sjNxXvgETXSctwfuMXZFmbeSE2

- a method that uses a public key as its identifier, directly incorporating cryptographic keys into the did
- fully self-contained, without any additional services required
- direct-key mapping allows straightforward verification of ownership
- easy to create: can be generated with standard key-creation mechanisms
- in the FIWARE DSC ideally used for identification of individuals or services(e.g. holders)

Decoding:

The id is the base58-btc encoded key, prefixed with the multicodec identifier of the key-type.

“zDn” as prefix of the ID indicates that it is a P-256 key.

did:web

did:web:one.batterypass.firmware.dev:did

- a method designed to be used in web-environments
- provides a way to create identifiers that are resolvable via standard web protocols
- the did-document is available at a well-known path, that can be created from the identifier
- in the FIWARE DSC, it could be used for participating organizations

Decoding:

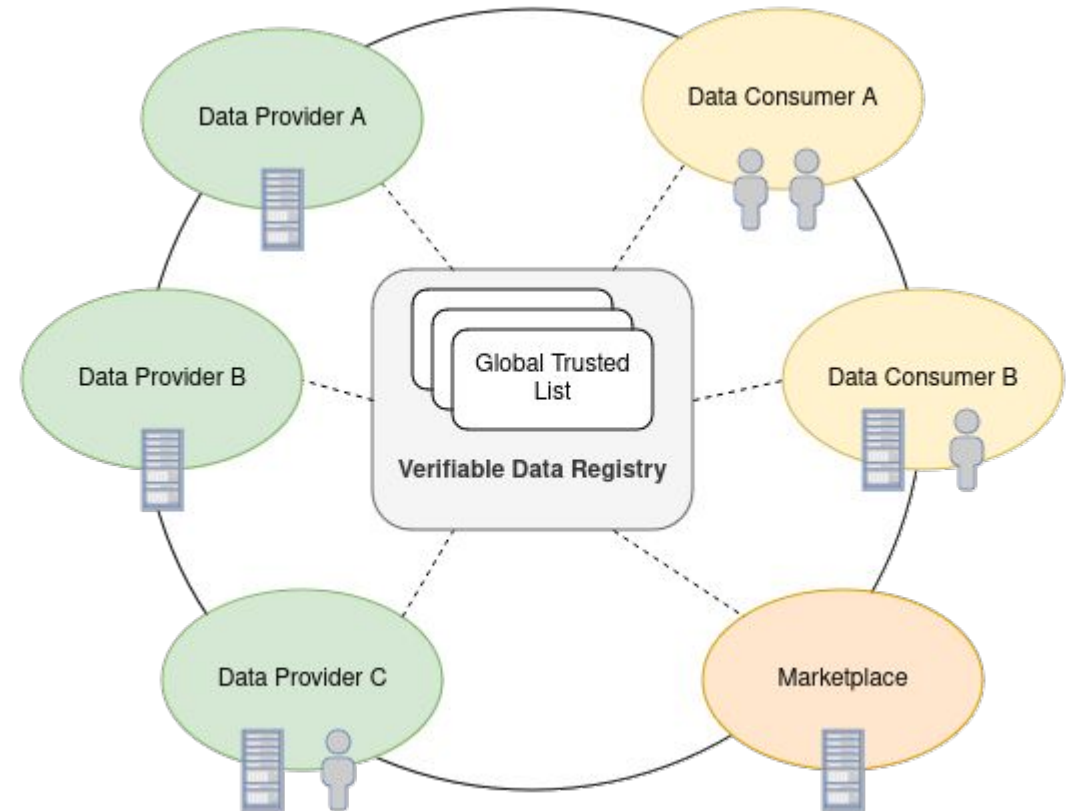
Following the method-definition, the id can be resolved to the document address:

<https://did-provider.dev.seamware.io/.well-known/did.json>

Https is enforced by the method and `did` as additional component provides information that the document is located under the sub-path `/did`.

Decentralized IAM in the FIWARE DSC

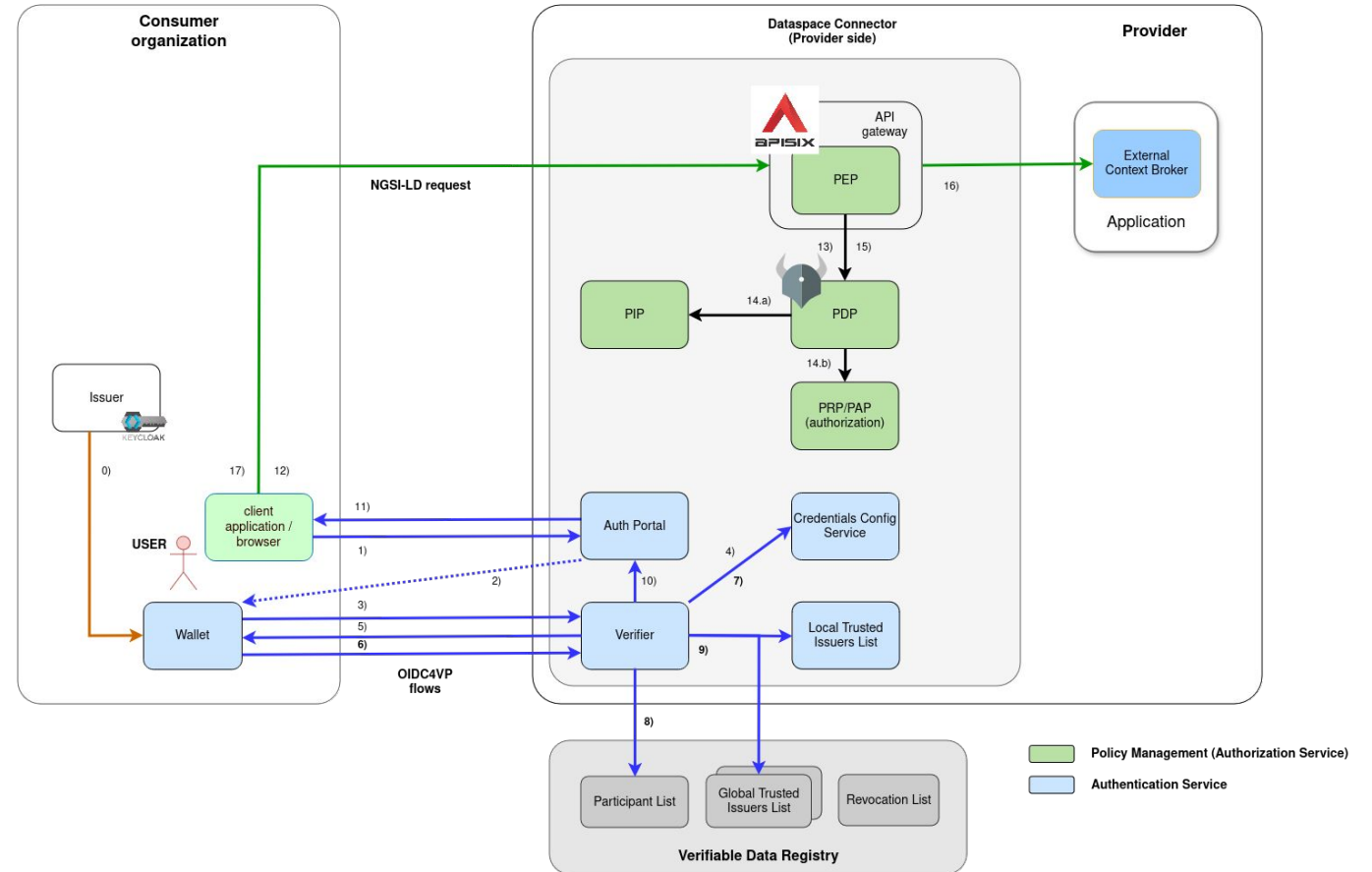
- Participants are registered in one/multiple Global Trusted Lists
- each participant can access the trusted list and check for registered participants
- additional Trusted Lists can be used, in order to incorporate public trust providers
- integration of GAIA-X Trust Framework could be achieved through this



Decentralized IAM in the FIWARE DSC

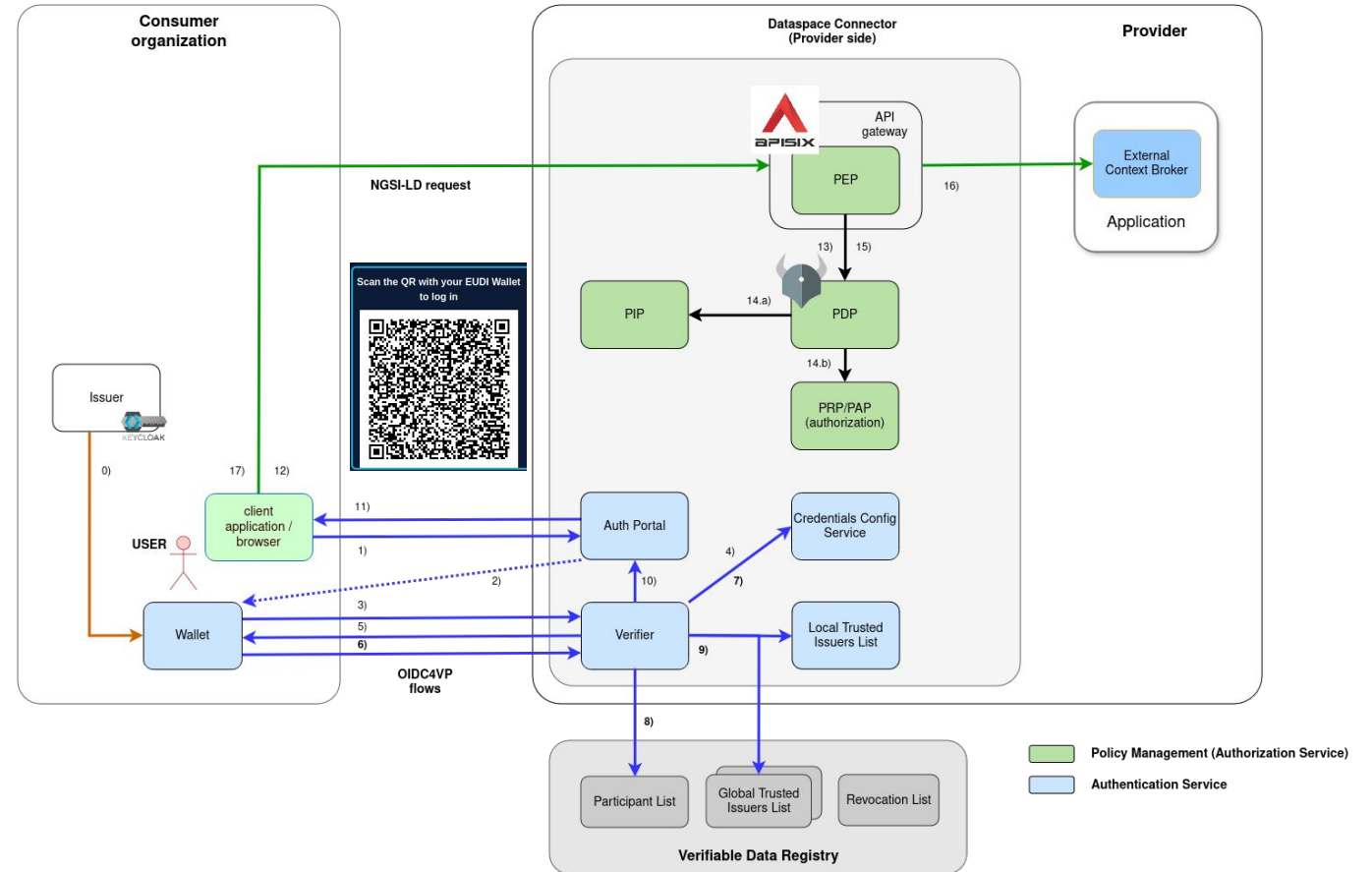
0. Issuance:

- Consumer Organization issues Verifiable Credential to its user
- Follows the [OID4VCI Standard](#)
- [Keycloak](#) is used as the default issuer in a FIWARE Data Space
- Credential will be stored in a [OID4VC](#) compliant wallet



Decentralized IAM in the FIWARE DSC

1. Users tries to access a protected resource and gets redirected to the Auth-Portal
2. Portal(as part of the Verifier) provides a QR with connection information for the Wallet
3. Wallet connects the Verifier with the provided information(f.e. service and scope) to start the OID4VP flow
4. [VCVerifier](#) requests the credentials configuration from the [Credential Config Service](#)
 - allow to define which credentials to be requested for a given service
 - defines which trusted lists to be used per credentials type/service



Authentication - Credentials Config

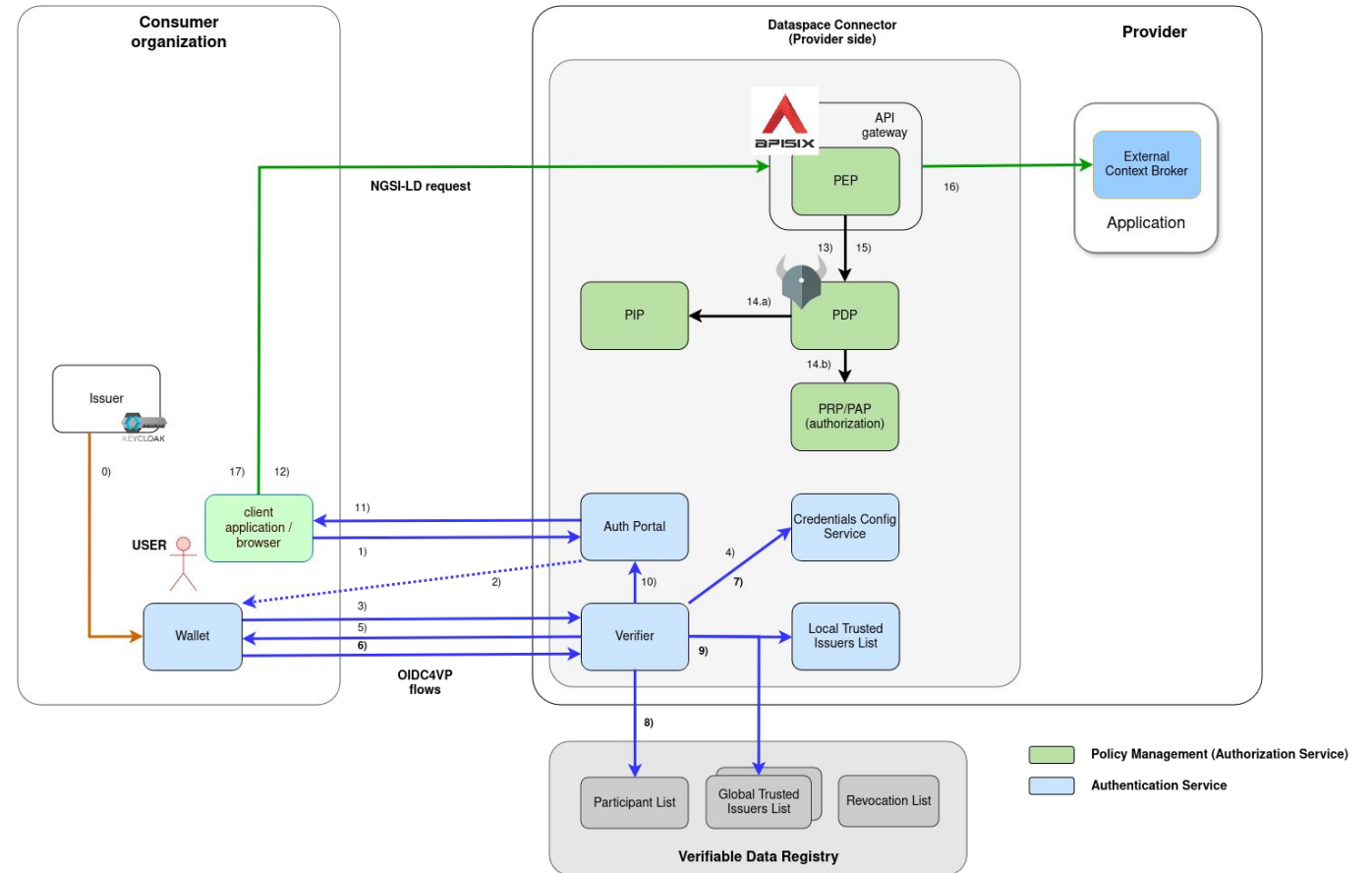
The required Credentials for a given Service and Scope can be fine-grained configured through the Credentials Config Service. It provides a REST-API to specify:

- the required Credentials
- the Trusted Lists to be used
- Scopes and their Configuration

```
{
  "id": "target-service",
  "defaultOidcScope": "read",
  "oidScopes": {
    "read": [{
      "type": "UserIdentityCredential",
      "trustedParticipantsList": ["https://my-dataspace.org"],
      "trustedIssuersList": ["https://my.default.svc.local"]
    }],
    "write": [{
      "type": "EmployeeCredential",
      "trustedParticipantsList": ["https://my-dataspace.org"],
      "trustedIssuersList": ["https://my.default.svc.local"]
    }],
  },
}
```

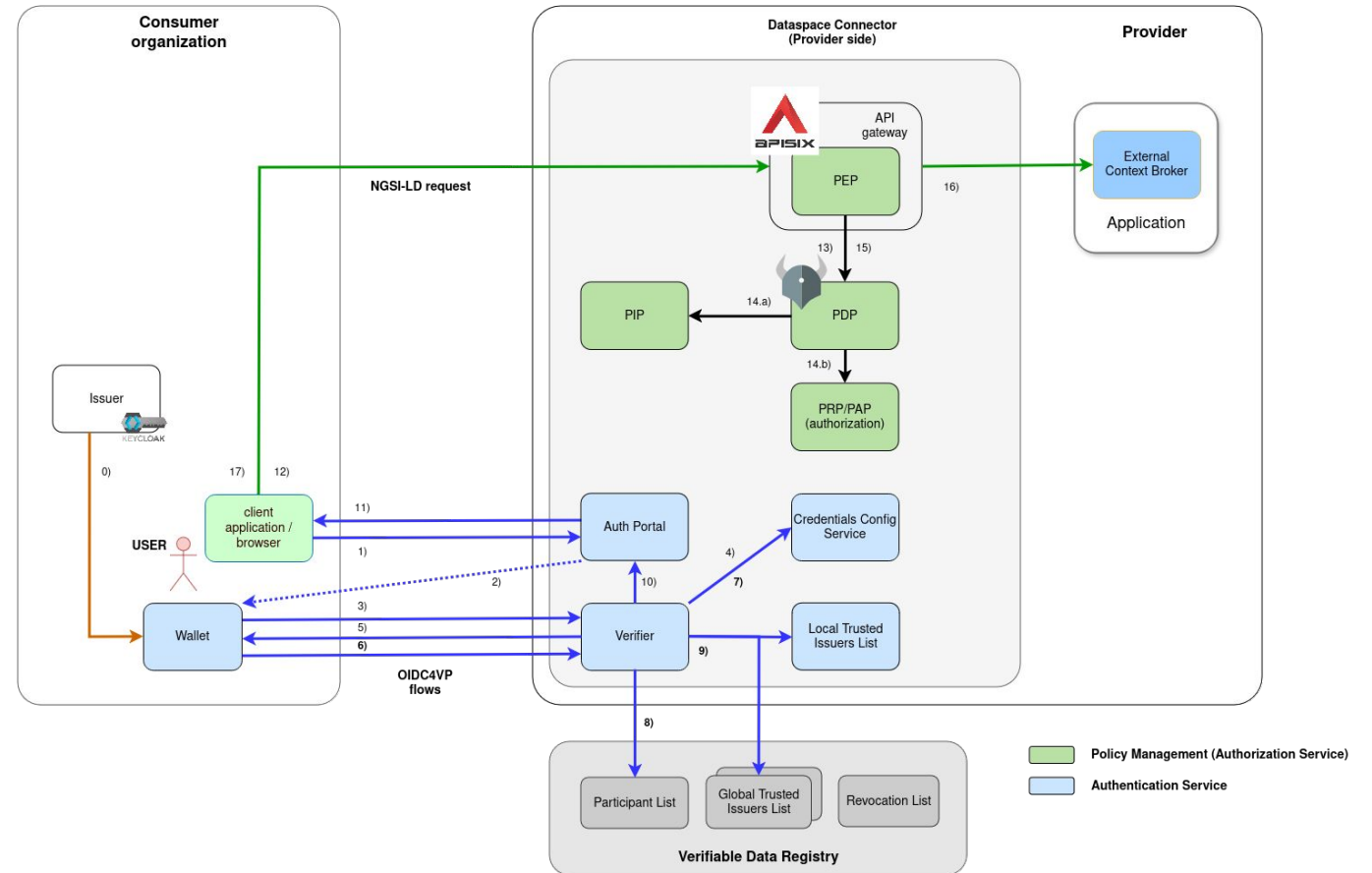
Decentralized IAM in the FIWARE DSC - Authentication

5. Verifier requests the credentials at the Wallet
6. Wallet creates the Verifiable Presentation and presents it to the Verifier
 - contains the requested Credentials
 - Signed with the Users Key
7. Verifier verifies the Signature of the Presentation and its contained credentials, checks at the Credentials Config Service if all Credentials are included and the Trusted Lists to be used.



Decentralized IAM in the FIWARE DSC - Authentication

8. Checks that the Credential was issued by a valid Data Space Participant
 - this is an optional step, that can be configured at the Credentials Config Service
9. Check the Credential at all configured internal or external Trusted Lists
 - the local [Trusted Issuers List](#) gives fine-grained control about the Claims to be trusted



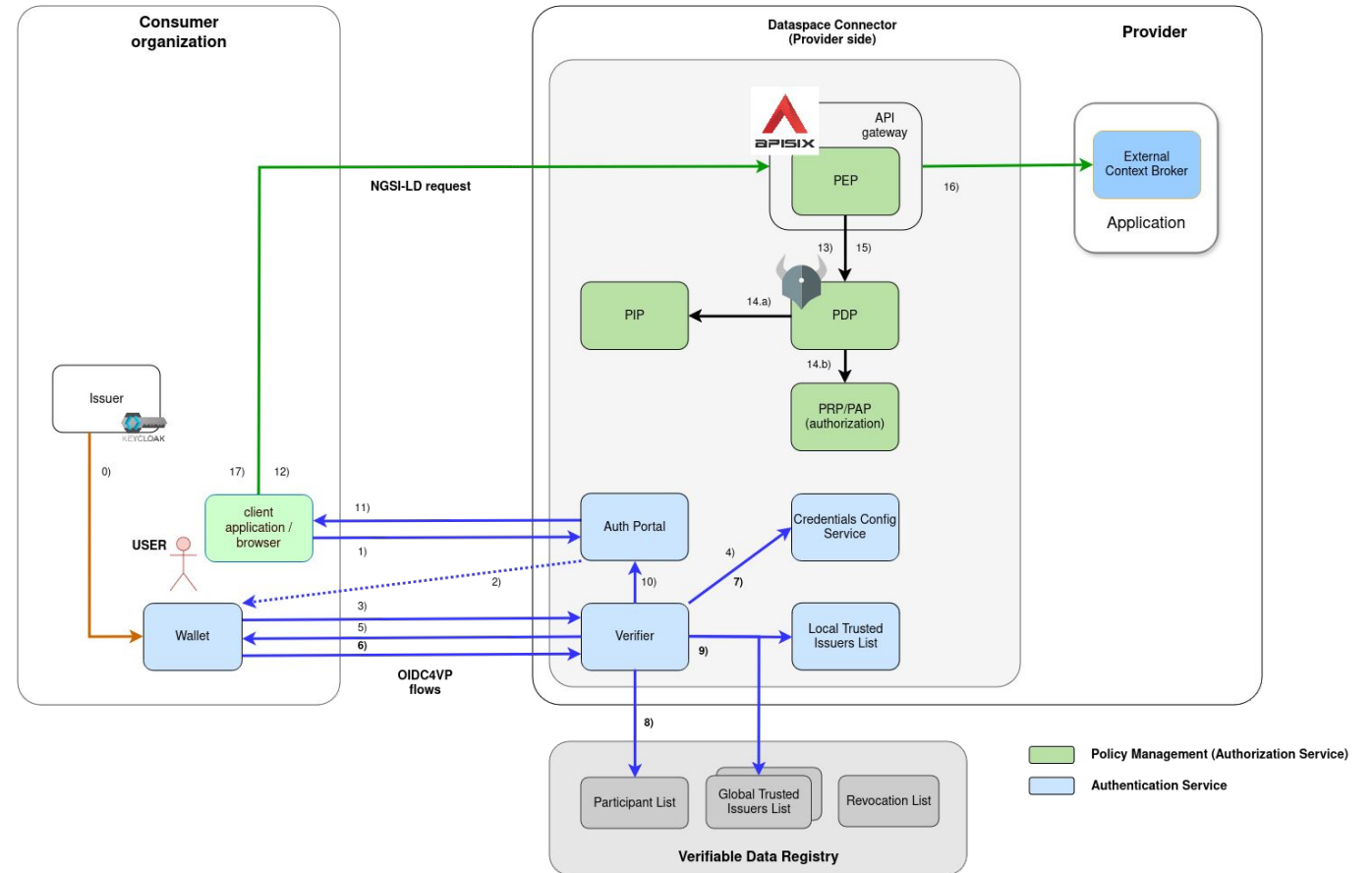
Authentication - Credentials Config

- implements the [EBSI Trusted Issuers Registry API](#) - [FIWARE Trusted Issuers List](#)
- in the ParticipantsList, only inclusion is checked
- IssuersList allows fine-grained control about trusted Credentials and Claims per Issuer
 - TimeRange that the configuration is valid
 - Types of Credentials the Issuer is trusted for
 - Claims and Values the Issuer is allowed to issue

```
{
  "did": "did:web:my-consumer.org",
  "credentials": [{
    "validFor": {
      "from": "2024-12-21:T17:00:00Z",
      "to": "2025-12-21:T17:00:00Z"
    },
    "credentialsType": "ConsumerCredential",
    "claims": [{
      "name": "roles",
      "allowedValues": [ "reader" ]
    }
  ]
}]
}
```

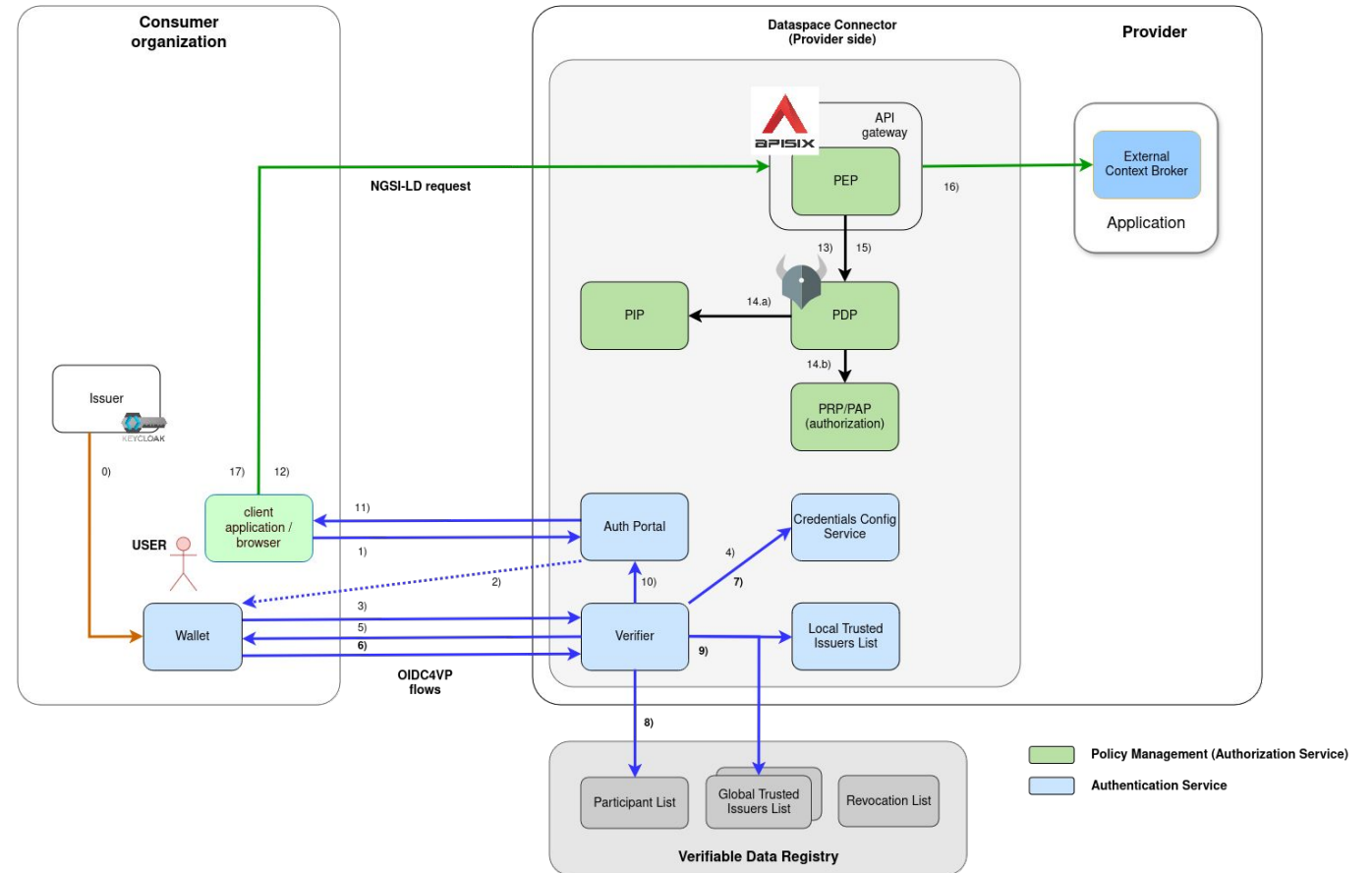
Decentralized IAM in the FIWARE DSC - Authentication

10. If verification is successful, a JWT Token, containing the credentials information is provided to the frontend
 - the JWT is signed with a Key generated by the Verifier
 - Public Key is provided through a .well-known JWKS endpoint
 - downstream components only have to verify the JWT, instead of the full Verifiable Credential
 - easier integration with “old” OIDC-based services
11. JWT is provided to initiating application



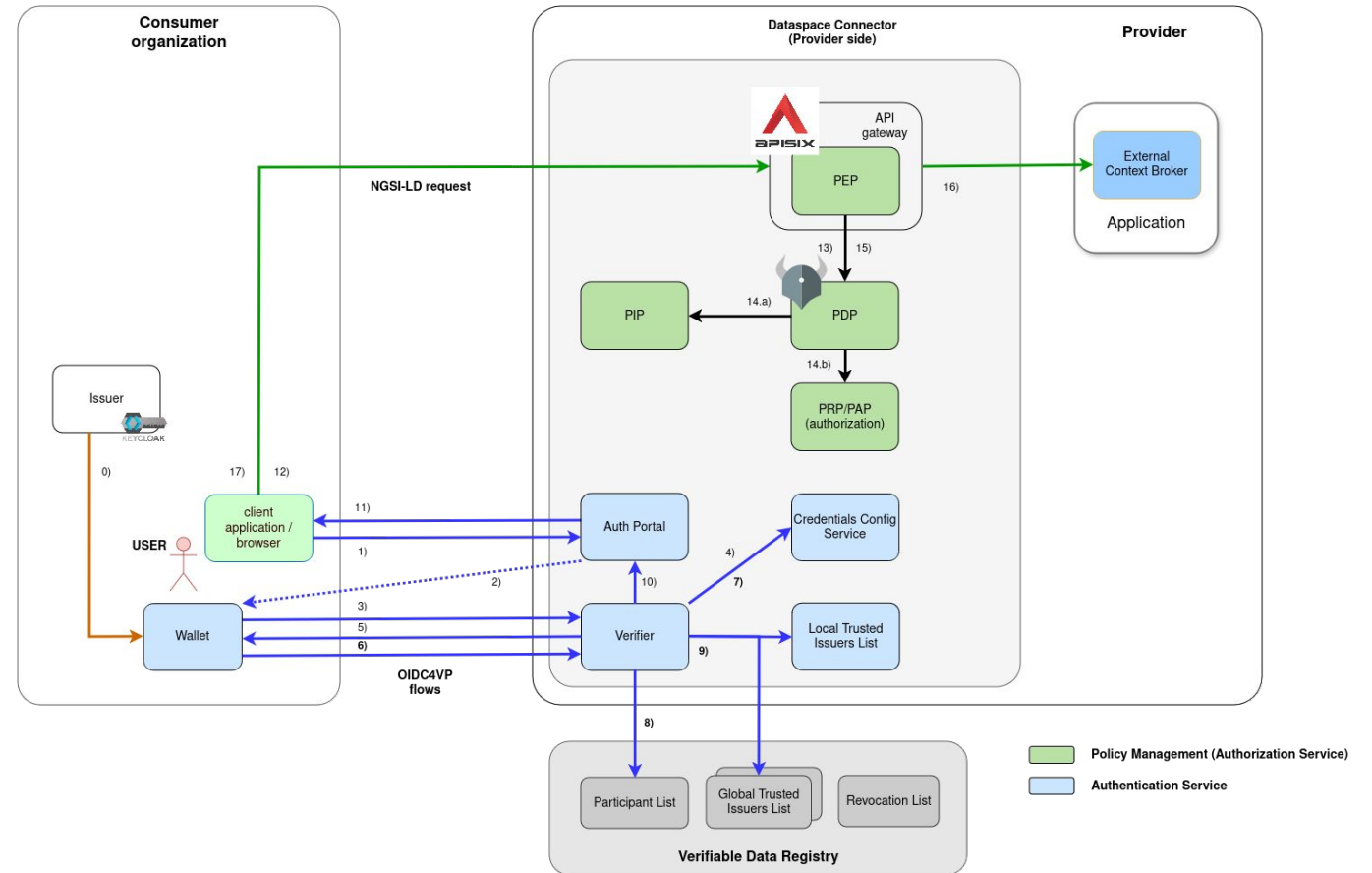
Decentralized IAM in the FIWARE DSC - Authorization

12. Application uses JWT to access the Provider Services through its Policy Enforcement Point(PEP)
 - [APISIX](#) is used as PEP
 - provides additional ApiGateway features
 - verifies the JWT at the Verifier and delegates the decisions to the Policy Decision Point(PDP)
13. PEP requests decision at the PDP
 - [Open Policy Agent](#) is used for policy evaluation
 - PEP provides request, request meta-data, authorization information and the request body



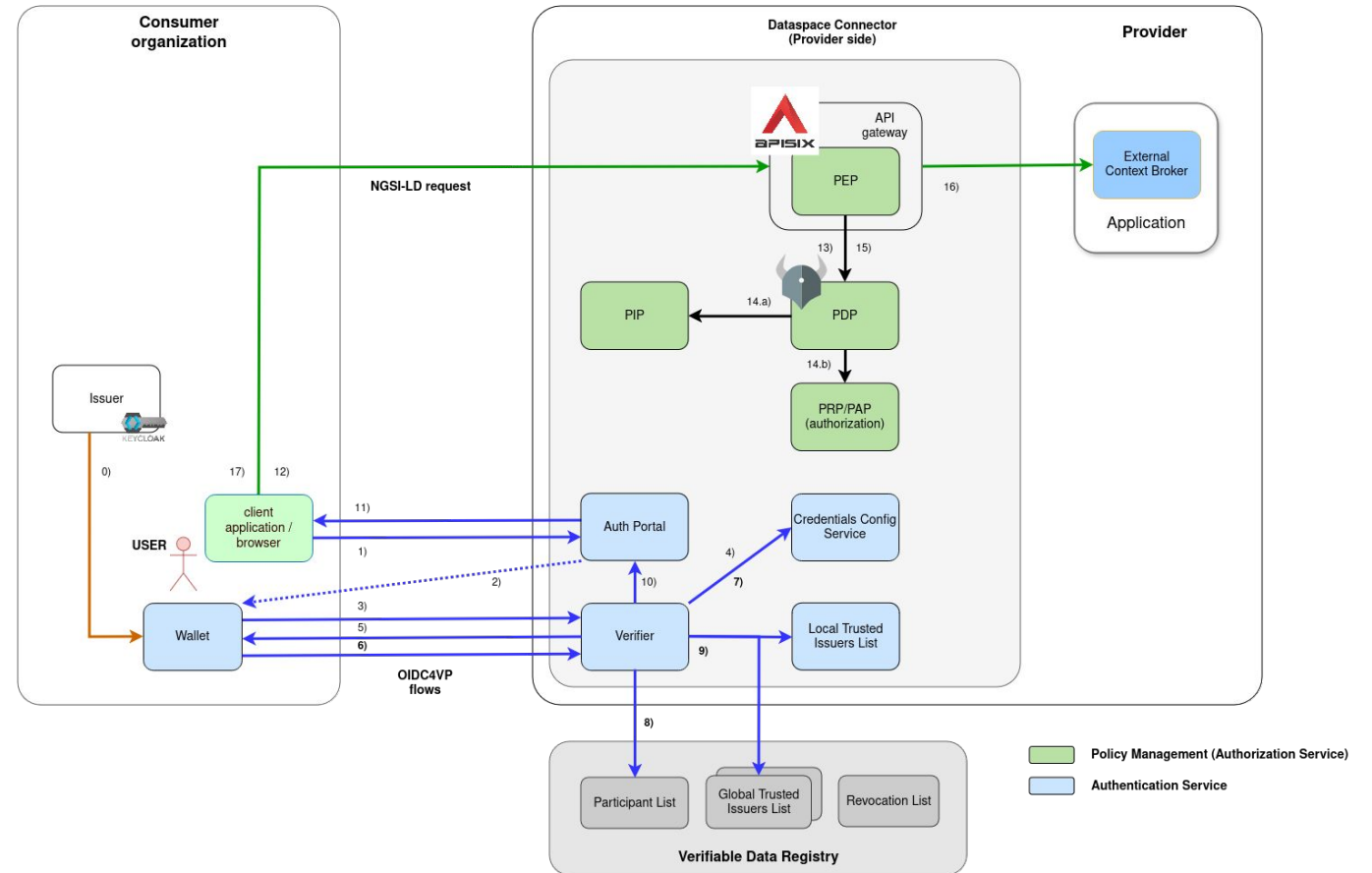
Decentralized IAM in the FIWARE DSC - Authorization

14. PDP evaluates the configured policies
- retrieves required information from the Policy Information Point(f.e. current time)
 - retrieves actual policies to be evaluated(happens asynchronous, sequential integrated for better understanding)
- Open Policy Agent is used due to its speed, stability and established community
 - already well integrated with APISIX(and various other ApiGateways)



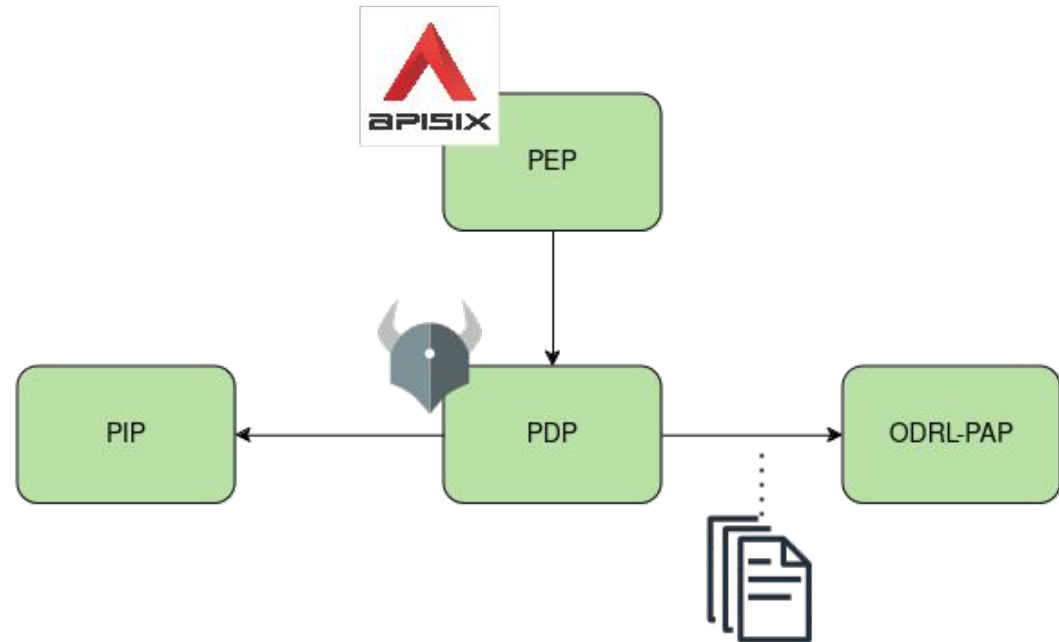
Decentralized IAM in the FIWARE DSC - Authorization

- 15. PDP returns decision to the PEP
- 16. PEP forwards the request to the target service
- 17. Response is provided to the requesting application



Decentralized IAM in the FIWARE DSC - Authorization

- handled by:
 - Policy Enforcement Point - [APISIX](#)
 - Policy Decision Point - [Open Policy Agent](#)
 - Policy Administration Point - [ODRL-PAP](#)
 - Policy Information Point
- Policies defined in ODRL, translated and enforced as Rego by OPA



ODRL

The Open Digital Rights Language(ODRL) is a standardized policy expression language used to represent permissions, prohibitions and obligations associated with digital content, enabling clear and enforceable rights management.

- Specified by W3C - [ODRL](#)
- allows to define policies on digital content
- Permissions can be used to express access-rights to resources
- possibility to couple them with resources - [DOME ODRL Profile](#)

ODRL

```
{
  "@context": {
    "odrl": "http://www.w3.org/ns/odrl/2/"
  },
  "@id":
  "urn:uuid:6ee8b922-d09a-4621-8ba1-46b6e811f682",
  "@type": "odrl:Policy",
  "odrl:permission": {
    "odrl:assigner": {
      "@id": "did:web:my-data-provider.org"
    },
    "odrl:target": "urn:ngsi-ld:entity:1",
    "odrl:assignee": {
      "@id": "did:web:my-data-consumer.org"
    },
    "odrl:action": {
      "@id": "odrl:read"
    }
  }
}
```

- @context provides the JSON-LD context for interpreting the policy structure
- @id is the unique identifier of the policy
- @type of the ODRL document, we only support odrl:Policy
- odrl:permission defines the actual rule:

*"did:web:my-data-provider.org allows
did:web:my-data-consumer.org to read the resource
urn:ngsi-ld:entity:1"*

ODRL to Rego translation

- [ODRL-PAP](#) provides CRUD-API for ODRL-Policies
- translates and stores policies
- policies are provided to [OPA](#) via the [Bundles-API](#)
- translation to Rego is configured through:
 - [mapping.json](#) connects policy entries with pre-defined Rego-Scripts
 - [rego-folder](#) contains all scripts provided for translation
 - can be extended and changed on deployment

Example:

"odrl:action": { "@id": "odrl:read" } is mapped to the rego-method "odrl_action.is_read(helper.http_part)"

- namespaces allow to define different behaviours for ODRL-Classes in the policies
- [helper methods](#) provide commonly used information, like http-headers, request-body or authorization information

Policy Enforcement

- every policy is condensed into a true/false decision
- all policies are combined to a single “main” policy - policy “grouping” by service is also allowed
- Open Policy Agent evaluates the main-policy by individually evaluating the single policies
 - allows on first policy evaluated to “true”
 - rejects if no policy evaluates to true

Main Policy:

```
package policy.main

import rego.v1
import data.policy.cdsoxrwbuc as cdsoxrwbuc
import data.policy.vmixqkzkuu as vmixqkzkuu
import data.policy.paofakgsoc as paofakgsoc
import data.policy.gvitaqhecj as gvitaqhecj
default allow := false

allow if cdsoxrwbuc.is_allowed
allow if vmixqkzkuu.is_allowed
allow if paofakgsoc.is_allowed
allow if gvitaqhecj.is_allowed
```

Examples

- restrict accessible targets - only entities of type “EnergyReport”

```
{
  "@context": {...},
  "@id": "urn:uuid:6ee8b922-d09a-4621-8ba1-46b6e811f682",
  "@type": "odrl:Policy",
  "odrl:permission": {
    "odrl:assigner": "did:web:my-data-provider.org",
    "odrl:target": {
      "@type": "odrl:AssetCollection",
      "odrl:source": "urn:asset",
      "odrl:refinement": [{
        "@type": "odrl:Constraint",
        "odrl:leftOperand": "ngsi-ld:entityType",
        "odrl:operator": "odrl:eq",
        "odrl:rightOperand": "EnergyReport"
      }],
      "odrl:assignee": "vc:any" ,
      "odrl:action": {
        "@id": "odrl:read"
      }
    }
  }
}
```

Examples

- restrict assignees - only assignees with the Role “Operator” and the credential type “OperatorCredential”

```
{
  ...,
  "odrl:permission": {
    "odrl:assignee": {
      "@type": "odrl:PartyCollection",
      "odrl:source": "urn:user",
      "odrl:refinement": {
        "@type": "odrl:LogicalConstraint",
        "odrl:and": [
          {
            "@type": "odrl:Constraint",
            "odrl:leftOperand": "vc:role",
            "odrl:operator": "odrl:hasPart",
            "odrl:rightOperand": {
              "@value": "OPERATOR",
              "@type": "xsd:string"
            }
          },
          {
            "@type": "odrl:Constraint",
            "odrl:leftOperand": "vc:type",
            "odrl:operator": "odrl:hasPart",
            "odrl:rightOperand": {
              "@value": "OperatorCredential",
              "@type": "xsd:string"
            }
          }
        ]
      }
    }
  }
}
```

Examples

- use(e.g. CRUD) is allowed between 31.12.2023 and 31.12.2024

```
{
  ...,
  "odrl:permission": {
    "odrl:target": "urn:ngsi-ld:data-entity:1",
    "odrl:assignee": "did:web:my-data-consumer.org",
    "odrl:constraint": [{
      "@type": "odrl:Constraint",
      "odrl:leftOperand": "odrl:dateTime",
      "odrl:operator": "odrl:gt",
      "odrl:rightOperand": {
        "@value": "2023-12-31",
        "@type": "xsd:date"
      }
    },
    {
      "@type": "odrl:Constraint",
      "odrl:leftOperand": "odrl:dateTime",
      "odrl:operator": "odrl:lt",
      "odrl:rightOperand": {
        "@value": "2024-12-31",
        "@type": "xsd:date"
      }
    }
  ],
  "odrl:action": "odrl:use"
}
```

Links

- Slides:
 - <https://github.com/wistefan/presentations>
- FIWARE Data Spaces:
 - <https://github.com/FIWARE/data-spaces>
- FIWARE Data Space Connector:
 - <https://github.com/FIWARE/data-space-connector>

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